

Caries Management: Diagnosis and Treatment Strategies

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Traditional caries management has consisted of the detection of caries lesions followed by immediate restoration. In other words, caries was managed primarily by restorative dentistry. However, when the dentist removes tooth structure with a handpiece, an irreversible process begins. Placing a restoration does not guarantee a sound future for the tooth; on the contrary, it may be the start of a restorative cycle in which the restoration will be replaced several times (Fig 5-1). This is reflected in the philosophy of Dr V. Kim Kutsch: "I get to the cause of the disease, instead of just treating the symptoms. When you're focused on just treating the symptoms surgically, which traditionally is what we've done in dentistry, you never catch up. You never get ahead."[1] Caries is like every other bacterial disease, and restoration without removal of the causative factors is like placing a new roof on a burning house.

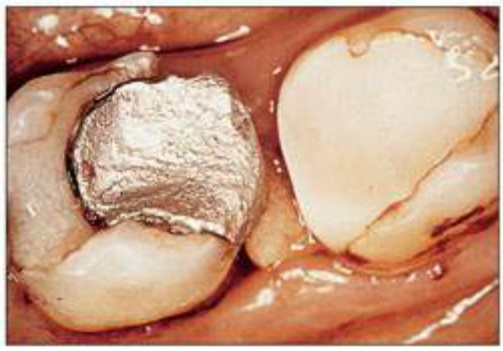


Fig 5-1 Iatrogenic damage caused by repeated treatment procedures.

A new paradigm for caries management, patient-centered caries management (PCCM), is recommended.[2] PCCM is enabled by the International Caries Detection and Assessment System (ICDAS) clinical visual scoring system for caries detection and activity.[2] When integrated into the caries management by risk assessment (CAMBRA),[3,4] treatment of caries is a process that can be controlled so that lesions may never form or, if they do, lesion progression can be arrested. The PCCM strategy involves a comprehensive assessment and includes evaluation of the caries risk and protective factors, detection and diagnosis of caries, and determination of the activity of existing caries in order to establish the individual patient's caries risk. The information collected from the comprehensive assessment is synthesized into a personalized treatment plan, composed of preventive and noninvasive treatment options, operative and minimally invasive treatment options, and maintenance care involving recall appointments for reassessment and monitoring (Fig 5-2). The treatment options recommended for specific caries lesions and patients will depend on a variety of prognostic factors, including lesion activity, and the monitoring of lesion behavior over time. It is important to mention that preventive regimens will arrest the caries process by redressing the imbalance between demineralization and remineralization.[5]

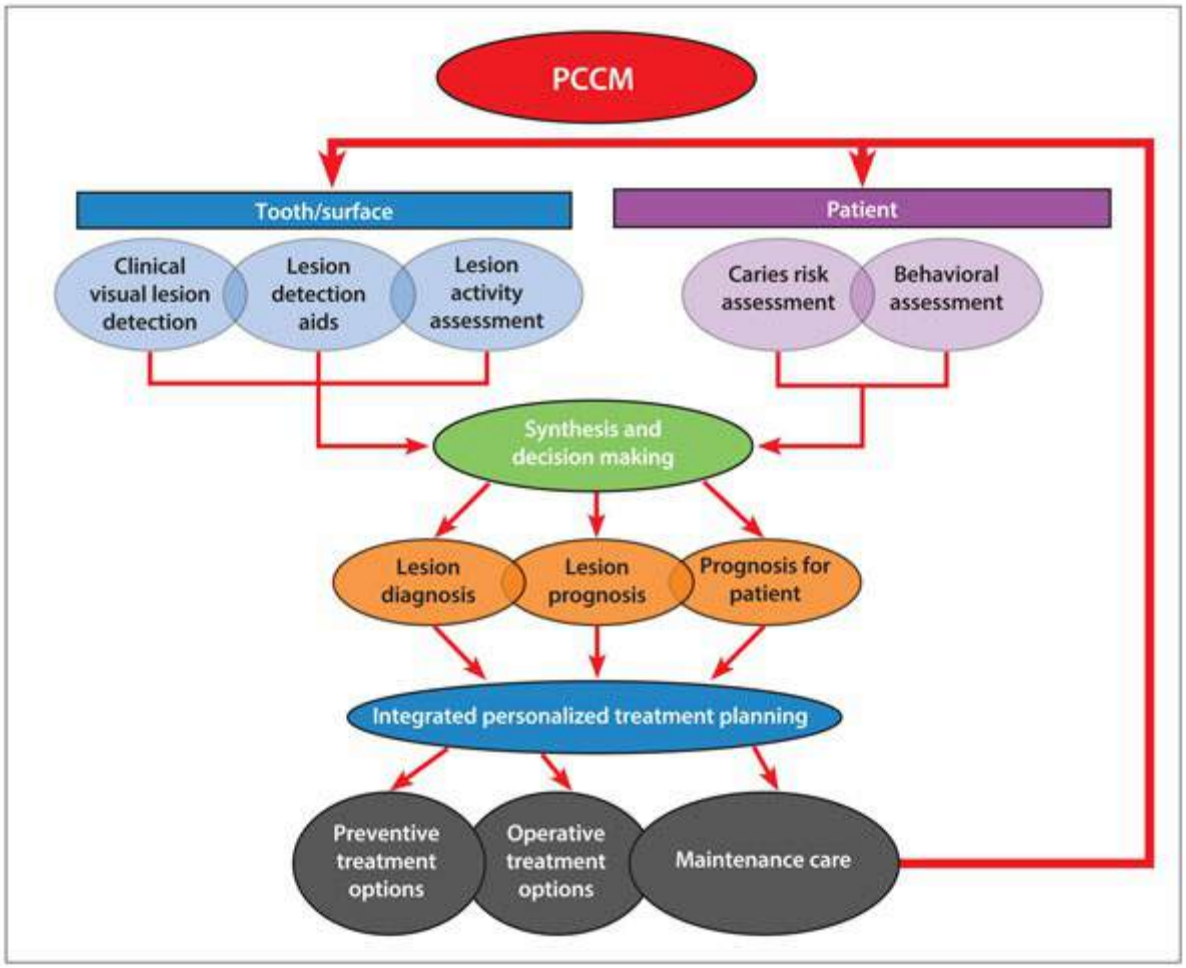


Fig 5-2 Algorithm for PCCM. (Modified from Pitts and Richards[2] with permission.)

The treatment goal in caries management should be to prevent new lesions from forming, to detect lesions sufficiently early in the process so that they can be treated and arrested by noninvasive means,[5] and to educate patients on the cause of the disease to gain their cooperation with recommended preventive strategies.

The Caries Process

Our understanding of the nature of any disease determines how we diagnose it and how we manage it. This general principle also applies to dental caries. Through scientific research, our understanding of the nature of caries and factors relating to its development has changed over time, as has the way we diagnose and manage the disease. The initiation of the development of dental caries by the concurrent action of three primary factors (the tooth, dental plaque, and diet) were presented in the 1960s in a model of overlapping circles.[6] Since then, the model has been

supplemented with factors that modulate the actions of the primary factors to determine the manifestation and clinical severity of caries[7] (Fig 5-3). At first sight, these circles constitute a simple model to explain caries risk, which is represented by the overlap of the three inner circles. When one of the risk factors increases, the respective circle becomes larger, as does the overlap of the circles, indicating increased caries risk.

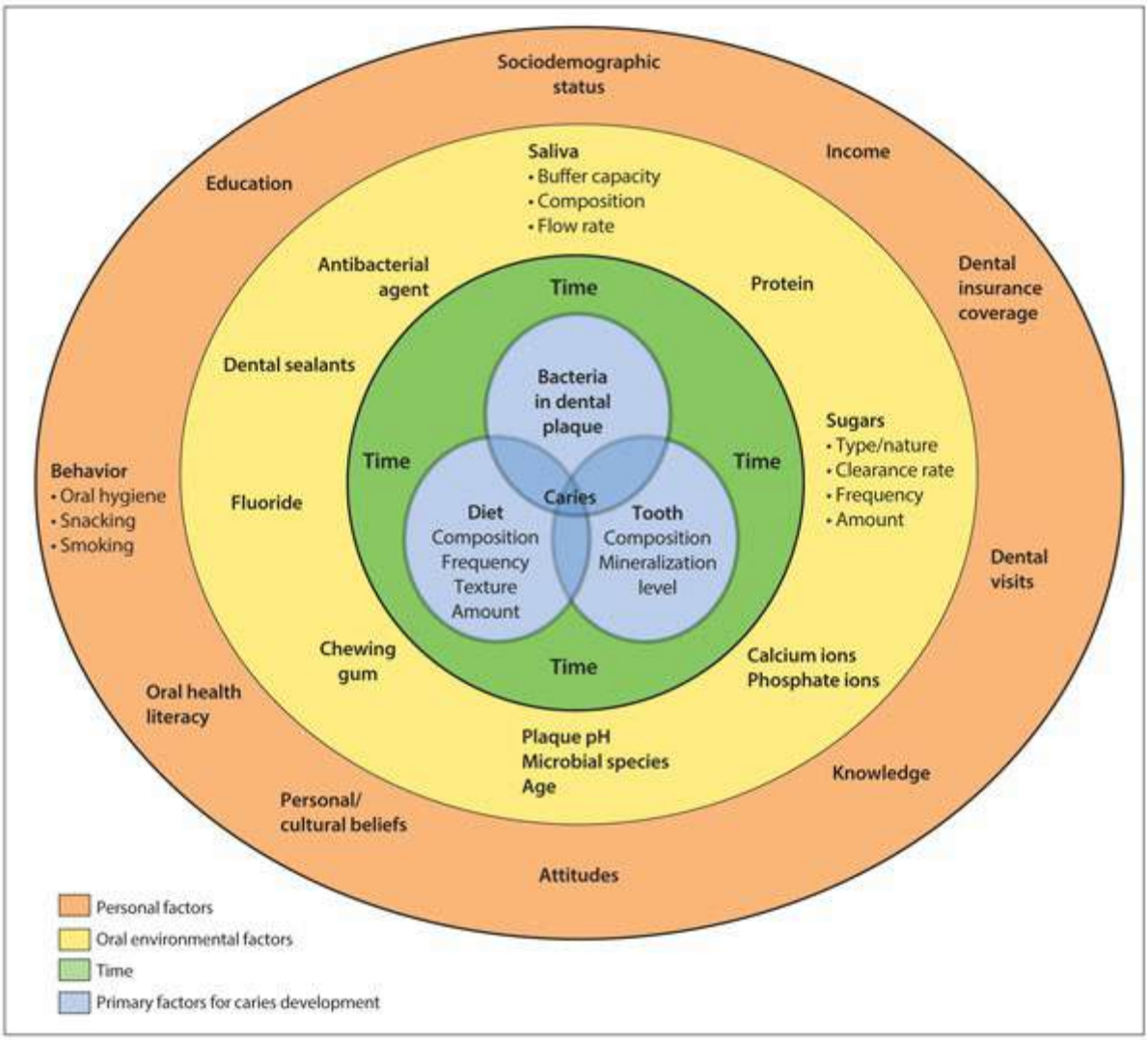


Fig 5-3 Factors influencing the equilibrium between the three prerequisites for the caries process as first described by Keyes and Jordan.[6] (Modified from Selwitz et al[7] with permission.)

Dental plaque

The prevalence of mutans streptococci and lactobacilli is associated with dental caries.[8–10] *Streptococcus mutans* is involved in caries lesion formation from its initiation, while lactobacilli flourish in a carious environment and contribute to caries progression. Dental plaque may be more cariogenic locally where mutans

streptococci and lactobacilli are concentrated, but in everyday practice it is difficult for the dentist to identify cariogenic plaque to make this knowledge useful in treating individual patients. Plaque can be sampled and mutans streptococci and lactobacilli levels quantified, but the procedure is quite complicated and requires the support of a microbiologic laboratory. It is easier to count mutans streptococci and lactobacilli in saliva, and kits are commercially available for this purpose. However, these counts do not give site-specific information and are poor predictors of high caries activity in general, although low counts or absence of mutans streptococci are good predictors of low caries activity.[11]

High numbers of mutans streptococci and lactobacilli are probably the consequence of a high sugar intake and the resulting periods of low pH levels in dental plaque.[12,13] Inversely, it has been shown that restriction of sugar intake reduces the numbers of mutans streptococci and lactobacilli.[12,14] In one study of Weight Watchers clients who complied with a non-cariogenic diet, the numbers of mutans streptococci and lactobacilli were reduced by half.[15] A comparable reduction was found in subjects who reduced their sugar intake frequency from 7.2 to ⅛ times per day.[16] However, the decrease in mutans streptococci was more pronounced on buccal than on proximal tooth surfaces, and, interestingly, the pH response to glucose was reduced in buccal but not in interdental plaque.[16,17] Six weeks after completing the sugar restriction period, the numbers of mutans streptococci increased again.[16]

The oral flora colonizes on teeth continuously, but it takes up to several days before the dental plaque contains enough acidogenic bacteria to lower plaque pH to the level that causes demineralization.[18] Theoretically, plaque removal every second day would be sufficient. If the dentition is professionally cleaned, an even lower frequency of home-care cleaning has been demonstrated to prevent caries.[19,20] But considering the caries prevalence in the prefluoride era, it is obvious that few people are capable of cleaning their teeth to a level adequate to prevent caries.

Teeth

Teeth consist of a calcium phosphate mineral that demineralizes when the environmental pH lowers. As the environmental pH recovers, dissolved calcium and phosphate can reprecipitate on remaining mineral crystals. This process is called *remineralization*. Remineralization is a slower process than demineralization. When remineralization is given enough time, it can eliminate the damage done during demineralization, but in the absence of this, the caries process will progress and a

lesion will develop. Dentin is more vulnerable than enamel because of structural differences and greater numbers of impurities in the crystal lattice, which facilitate its degradation in an acidic environment. For many years, much emphasis was given to the preeruptive effect of fluoride improving the quality of the dental hard tissues. However, it is now clear that posteruptive use of fluoride is far more protective against caries.[21–23] Enamel in primary teeth is less mineralized and more variable than permanent enamel. Optically it is more opaque, which may represent greater porosity. It gives the clinical impression of wearing more quickly and being less resistant to caries.[24]

Diet

Dietary carbohydrates are necessary for the bacteria to produce the acids that initiate demineralization. In general, dietary advice for caries prevention is based on three principles: *(1)* the drop in pH lasts for approximately 30 minutes; *(2)* the frequency of intake is more important than the quantity; and *(3)* the stickiness of foods is an important factor in their cariogenicity. It has become obvious, however, from many epidemiologic studies where fluoride is used daily, that sugar consumption and caries prevalence have become less tightly related for many individuals. Even when there was a significant correlation between sugar consumption and caries prevalence, the caries-preventive effect of sugar restriction was small. For instance, in Basel, Switzerland, wartime restriction reduced sugar supply from about 48 to 16 kg per person per year, but the number of caries-free children rose only from approximately 3% to 15%.[25,26] At that time, the improvement seemed impressive, but it was dwarfed by the effect of nationwide optimum fluoride administration, when the number of caries-free children rose from approximately 7% to more than 50%, with a reduction of sugar supplies from approximately 42 to 39 kg per person per year.[25] Obviously, fluoride administration was far more effective in reducing caries than lowering sugar supplies. With this evidence, the role of dietary counseling in caries prevention should be reexamined. This does not negate the value of diet analysis and advice for patients presenting with multiple caries lesions, but the importance of the proper use of fluoride should always be emphasized.

Information gathered with the reliable pH-telemetry method has revealed that a pH drop induced by eating may last for hours if there is no stimulation of the salivary flow.[18,27,28] Even the consumption of an apple can depress the pH for 2 hours or longer.[18] Long pH depressions will be most prevalent in areas where saliva has little or no access, and these areas are the most caries prone. It is unknown how

much additional harm is caused by a second sugar intake during such a period of low pH or how beneficial it is to omit a second sugar intake during that period. Foods believed to be "good" for teeth may not be better than foods that are supposedly "bad." A chocolate and caramel bar might be considered bad because it feels sticky. In reality, however, the caramel dissolves and leaves the mouth relatively quickly, whereas potato chips, generally considered less harmful, take a longer time to clear from the mouth.[29] During this retention, the carbohydrate fraction may be hydrolyzed to simple sugars, providing a substrate for the acidogenic bacteria.[30]

All the uncertainties about the determinants of the cariogenicity of foods make it impossible to provide strict dietary guidelines. To snack in moderation, limited to 3 or 4 snacks a day, is the only wise recommendation.

Time

Time affects the caries process in several ways. When caries was considered to be a chronic disease, time was introduced to indicate that the substrate (dietary sugars) must be present for a sufficient length of time to cause demineralization.[31] Now we know that caries is not a chronic disease and that its effects can be arrested or completely repaired if enough time is given for remineralization. Finally, it is clear that caries lesions do not develop overnight but take time; in fact, it may take years for a caries lesion to develop a distinct discontinuity or break in its surface integrity (otherwise called *cavitation*). This potentially gives the dentist and the patient ample time for preventive treatment strategies.

Fluoride

Experiments have shown that fluoride protects enamel more effectively when it is present in the ambient solution during acid challenges than when it is incorporated into the enamel crystal lattice.[21] One of the mechanisms by which fluoride inhibits demineralization is by reprecipitation of dissolved calcium and phosphate, which prevents these constituents from being leached out of the enamel into the plaque and saliva.[23] Part of the reprecipitation takes place at the surface of the tooth. This narrows the pores in the enamel surface that provide diffusion pathways for the acids produced in the dental plaque to penetrate into the enamel. Acid penetration is thus hampered. In addition, during periods in which the ambient pH is higher than 5.5, fluoride will facilitate remineralization, promoting lesion arrest and repair. A lack of fluoride constitutes a caries risk.

The retention of fluoride in the mouth is site specific. In plaque and saliva and in dentin samples fixed in dental splints, most fluoride was found on the labial surfaces of maxillary incisors and buccal surfaces in the mandibular molar region after rinsing with a fluoride solution.[32,33] In addition, it was observed that fluoride from passively dissolving fluoride tablets remained highly concentrated only at the site of tablet dissolution. There was very little or no transport of fluoride between the right and the left sides of the mouth and between the maxillary and mandibular arches.[33] Because of this, localized caries lesions in the mouth may be related to an insufficient spread of fluoride when subjects use fluoride toothpaste. When patients use fluoride toothpaste, they should be encouraged to spit out any excess rather than to rinse vigorously with water. Alternatively, the toothpaste slurry may first be used as a mouth rinse.

Saliva

The important role of saliva is clearly demonstrated by the rampant caries (Fig 5-4) that may occur in subjects with compromised salivary flow. These subjects lack the protective qualities of saliva (Table 5-1), of which the flow rate and buffering capacity may be the most important. Both help to neutralize and clear the acids and carbohydrates from dental plaque. Clearance, however, is not uniform throughout the mouth and may be slowest at the labial aspects of the maxillary incisors and buccal aspects of the mandibular molars. Other sites in the dentition may not be easily accessible to saliva as a result of an individual's anatomy, including interproximal spaces and fissures. Dental plaque in a cavitated area of a tooth may also be protected from salivary clearance.



Fig 5-4 Rampant caries. (Courtesy of Claudia Isabel Contreras, San Antonio, Texas.)

Table 5-1	Functions of components of saliva involved in the caries process

Components	Functions
Mucins, lysozyme, lactoferrin, lactoperoxidase	Antibacterial
Histamine, agglutinin, cystatines	Antibacterial
Secretory immunoglobulin A, alphaamylase, histatins, histidine-rich proteins	Antibacterial
Proline-rich glycoprotein, mucins	Lubrication/viscoelasticity
Mucins	Inhibition of demineralization
Proline-rich proteins, statherin, calcium phosphate	Remineralization
Bicarbonate, phosphate, urea, arginine rich proteins	Buffering

The sites that are difficult for saliva to reach also may be difficult to reach with mechanical cleaning devices, such as a toothbrush or dental floss. Plaque and food may adhere for long periods of time in these areas, making these sites more prone to caries. Furthermore, this caries risk factor may be overlooked in children, whose teeth appear to be clean as judged from the sites that are easily cleaned. These children may brush twice daily, have only a moderate frequency of sugar intake per day, and still develop caries lesions at these sites. The most feasible way to prevent caries lesions at these sites is by thorough oral hygiene measures and use of a fluoride-containing toothpaste so that plaque is removed and fluoride is applied. In Fig 5-5a, a white spot lesion had developed on an erupting permanent second molar because of insufficient toothbrushing. However, the progression of the lesion was stopped by a better understanding of how, when, and why to brush; keeping the toothbrush at a right angle in the fissure; and using a toothpaste with fluoride. The arrested lesion is shown 1 year later (Fig 5-5b).

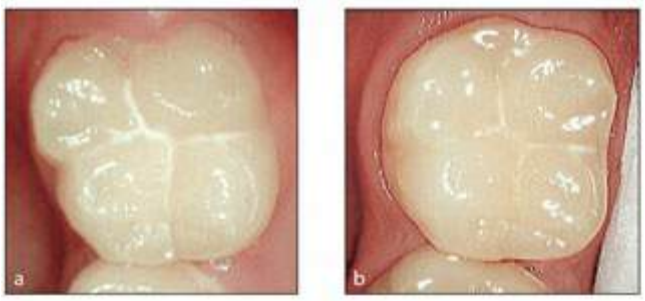


Fig 5-5*(a)* A permanent mandibular second molar with an active white spot lesion in the occlusal fissures. *(b)* The same lesion after 1 year of brushing with toothpaste containing fluoride. (Courtesy of Jette Christiansen, Copenhagen, Denmark.)

Social and demographic factors

Many studies have shown that, at least in the western world, dental caries is more prevalent in lower socioeconomic categories, in less affluent areas, and among some ethnic minorities[34] (Table 5-2). Differences related to socioeconomic status are very clear for the primary dentition and less clear for the permanent dentition, although this pattern may differ in other parts of the world. Studies have shown that for the prediction of caries lesion development, social and demographic factors may be successful in very young children without a long dental history, but for older children, clinical parameters such as the dmfs (decayed/missing/filled surfaces in primary teeth) and DMFS (decayed/missing/filled surfaces in permanent teeth) as shown in Table 5-3 are more predictive.[11,35–37] In the elderly population, however, root caries lesion development again seems to be more prevalent in people from lower socioeconomic backgrounds.

Table 5-2 **Percentages of 6- and 12-year-old caries-free children and mean dmfs and DMFS scores in children who were not caries free in 1989, 1993, and 1996, according to socioeconomic level[34]**

	6-year-old children						12-year-old children				
Socioeconomic status	**%**	**d**	**m**	**f**	**dmf ± SD**		**%**	**D**	**M**	**F**	**DMF ± SD**
Low											
1989	43	4.7	0.6	2.8	8.2 ± 8.5		46	0.2	0.0	3.4	3.6 ± 2.2
1993	39	4.8	0.3	2.7	7.8 ± 9.0		67	0.1	0.0	3.9	4.0 ± 3.1
1996	49	5.0	1.1	3.5	9.6 ± 9.8		50	0.1	0.0	3.1	3.2 ± 2.4
Medium											
1989	60	3.4	0.8	1.3	5.5 ± 6.9		49	0.2	0.1	3.5	3.9 ± 2.7
1993	69	4.0	0.3	1.4	5.8 ± 7.5		61	0.3	0.0	2.5	2.8 ± 1.9
1996	79	3.7	0.9	0.4	4.9 ± 5.3		89	0.0	0.0	3.6	3.6 ± 2.6
High											
					2.1 ±						2.9 ±

1989	77	1.5	0.0	0.6	1.3		59	0.9	0.2	1.8	2.0
1993	77	2.5	0.2	1.7	4.4 ± 3.5		63	0.3	0.0	1.7	2.1 ± 1.6
1996	84	1.6	0.9	2.2	4.7 ± 5.1		86	0.3	0.0	2.0	2.3 ± 2.1

dmfs–decayed/missing/filled surfaces in primary teeth; DMFS—decayed/missing/filled surfaces in permanent teeth, SD–standard deviation.

Table 5-3	**Timeline of strongest clinical predictors of caries incidence**[11]						
	Age (y)						
	0 and 1	**2–5**	**6–9**	**10–13**	**14–21**	**22–45**	
Dentition	Primary	Primary	Mixed	Mixed	Early permanent	Mature permanent	M pe an re
Predictor*	Mutans streptococci	**dmfs, especially primary incisors;** mutans streptococci and lactobacilli	**dmfs, especially primary molars;** first molar occlusal morphology: DMFS	**DMFS, especially permanent first molars;** first molar occlusal morphology; incipient smoothsurface lesions	**Incipient smoothsurface lesions;** DMFS	Not studied	C an DI nu te pe dis

*The strongest predictors are in bold. dmfs–decayed/missing/filled surfaces in primary teeth; DMFS–decayed/missing/filled surfaces in permanent teeth.

Caries prediction

A good estimate of the caries risk is necessary to make the most appropriate treatment decisions. A recent study[38] described developmental trajectories of caries experience in the permanent dentition to age 32 years. Three caries-experience trajectories were identified and categorized as *high* (approximately 15% of the population), *medium* (approximately 43% of the population), and *low* (approximately 42% of the population) DMFS. All were relatively linear, although the higher trajectories were more S-shaped. The data suggest that, among individuals following a similar caries trajectory, caries rate is relatively constant across time. The trajectories already diverged at an early age of the individuals. This study

supports that indicators of past caries experience are the strongest predictors of future caries, and the status of the most recently erupted or exposed surfaces is a strong predictor for the newly emerging surfaces. One very elegant review plots the strongest clinical predictors against age and dentition[11] (see Table 5-3).The sensitivity and specificity of the predictors varied from 0.60 to 0.80, indicating that 20% to 40% of the children who developed caries lesions during the study were not identified as being at high risk for caries and that 20% to 40% of the children who were not caries active were, in contrast, predicted to develop caries lesions. In addition, it is important to recognize that none of the studies in the review were designed to predict the progression rate of carious demineralization at a specific site. So, even if it could be predicted that an individual would develop caries lesions, the lesion that is going to progress cannot be confidently identified.

Rate of caries lesion progression

The decline in caries prevalence has been accompanied by a change in lesion behavior. Caries lesions progress more slowly than they did several decades ago, probably due to increased use of fluoride, which delays lesion progression. It is clear that the progression rate is not the same for each site. Little is known about the progression rate of fissure caries lesions. Longitudinal epidemiologic data from the 1950s, when fluoride was not yet widely used, showed that it took approximately 1 year for an enamel fissure lesion to develop into a dentinal lesion.[39] More recent data from the same geographic area, after the introduction of fluoride toothpaste, showed that 50% of the enamel fissure lesions had progressed to involve dentin within 2 years while 75% had become dentinal lesions after 4 years.[39] Data from Switzerland showed that, for permanent first molars, approximately 10%, 20%, and 45%, respectively, of sound fissures, fissures with a yellowish discoloration, and fissures with a dark discoloration progressed to carious dentin or tooth restoration in 4 years.[40] These progression rates were slightly lower than those found by the same investigators 10 years earlier. It is the experience of many clinicians that not all initial fissure lesions will develop to involve the dentin but that a number of them may become arrested. In populations in which not all fissure lesions develop into open cavities, the dentist will be confronted with clinically undetected carious dentin at the base of occlusal fissures.[41] This is not a new phenomenon; it was discussed as early as 1931.[42] Today, reported progression rates of pit and fissure caries lesions vary greatly.[43,44]

Based on epidemiologic data from the 1950s, the progression rates in the

permanent dentition of proximal caries lesions from initial enamel lesions to involving dentin was estimated to be 2 years at age 7 years and approximately 4 years at age 12 years.[37] Data collected after fluoride supplementation became available showed a progression rate of 3 to 4 years for proximal caries lesions to reach the dentin in 12-year-olds.[44] Shwartz et al[45] concluded from data collected in Sweden and the United States that it takes an average of 4 years for caries lesions to progress through the proximal enamel of permanent teeth. The progression rate seemed to be independent of the DMFS of the individuals.

Not all caries lesions progress, however. In one study, more than 50% of initial proximal caries lesions in 13-year-old adolescents had not advanced during a 3-year period.[46] The majority of the lesions that progressed were found in the adolescents who had the highest number of lesions at the start of the study, which probably reflects the difference between caries-active and caries-inactive individuals. Recently, Mejàre et al[47] published data on the caries lesion prevalence on the proximal surfaces of the posterior teeth at various ages (Fig 5-6). The prevalence was low, with the distal surfaces of the first molars being the most prone to developing caries lesions.

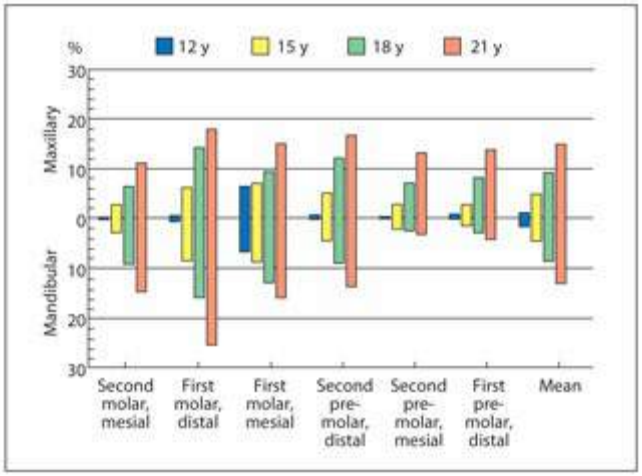


Fig 5-6 Prevalence (percentage of approximal surfaces affected) of dentinal lesions in the respective approximal surfaces in different age groups. (Data from Mejàre et al.[47])

Caries lesions on free smooth surfaces seem to progress more slowly than on proximal surfaces or in fissures. Many lesions do not progress into the dentin and even show regression to sound enamel.[37,48–50] In one study, dentists were asked at what point they would restore small noncavitated lesions on the buccal surfaces of teeth. Approximately 40% indicated that they would use a preventive rather than an operative strategy.[51] This indicates that many dentists believe that they are well able to judge the severity of buccal lesions and to monitor lesion development.

Altogether, the evidence indicates that the decline in caries prevalence has been accompanied by a decline in caries lesion progression rates. Between the initiation of caries and the involvement of dentin in the caries process, there is ample time for a preventive management strategy. This implies that the early lesion should be detected so that preventive treatment can arrest its progress and bring about lesion arrest/remineralization. If this strategy is successful, operative intervention will not be required.

Detection and Diagnosis

Diagnosing a caries lesion as a deviation from the normal appearance of enamel or root surface involves integration of the information obtained by clinical examination of the oral cavity, use of caries diagnostic aids, conversation with the patient, and biologic knowledge of the caries process.[52] Unfortunately, this process is often referred to as *caries detection*, which is simply an objective method of determining whether or not clinical signs of caries are present at one point in time.[53] Caries, however, is a highly dynamic process characterized by alternating periods of demineralization and remineralization. Caries begins and progresses when demineralization outgains remineralization, which is seen in a frequent eating condition in which the pH of the oral environment is never allowed to come to neutrality, or in inadequate use of fluorides. While detection only describes the clinical signs of caries, diagnosis tries to estimate the dynamics and determinants of the process. White spot lesion is found by detection; after diagnosis, it may be called an *initial caries lesion*.

Detection

The primary objective of early detection of caries is to limit the progression and impact of the disease after onset at as early a stage as possible. This objective is based on the fact that caries is reversible as well as preventable. Restoration of a cavitated caries lesion is costly in terms of time, resources, and oral health; therefore, a major goal is the prevention of demineralization and promotion of remineralization of early caries. This goal can be achieved only if the early clinical signs of caries are detected and effective prophylactic intervention is employed to reverse or arrest the lesion before the need for restorative intervention. The conventional caries detection methods are visual and radiographic examinations.

Visual examination

The first clinical sign of caries is a chalky and matte (rough) whitish surface (hence *white spot lesion*) (Fig 5-7). Microscopically, this white spot is a porous surface that can easily be stained into brown or black discoloration by chromogens from foods; thus, a caries lesion can be seen either as a white or as a brown/black spot lesion (Fig 5-8). There are noncavitated and cavitated stages of dental caries that can be identified and described using clinical signs.

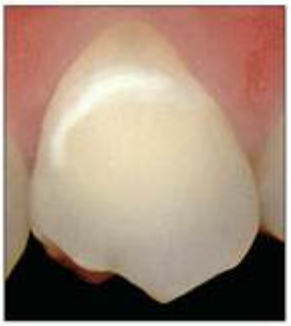


Fig 5-7 The first clinical sign of caries: a chalky and matte whitish surface.



Fig 5-8 White spot lesion discolored by staining has turned into a brown spot lesion.

Although it is the most common method used by clinicians, the detection of early caries by visual examination is considered subjective, less sensitive, and of poor specificity because of the various factors that might influence it. The presence of saliva, ambient light reflections, and bacterial plaque on tooth surfaces lowers the accuracy of visual examination, and the factors that affect color perception and interpretation render visual examination subjective. The sensitivity and specificity of this method of examination can be improved by certain prerequisites.

- *Clean tooth.* Most often the tooth is covered by a film of bacterial plaque that can camouflage a suspect lesion (Fig 5-9). It is necessary before caries detection that the plaque be removed gently with the explorer; detection may also be carried out immediately following prophylactic cleaning.



Fig 5-9 Reliable detection of caries lesions can be obstructed by plaque. *(a)* Plaque covers the labial surfaces and conceals a cavity in the maxillary right central incisor. *(b)* The same surface, cleaned, reveals the cavitated lesion. (Courtesy of K. R. Ekstrand, Copenhagen, Denmark.)

- *Dry tooth.* The teeth should be thoroughly dried with an air-water syringe (> 5 seconds) before all surfaces are carefully examined. The presence of saliva on tooth surfaces interferes with the detection of white spot lesions because of optical phenomena resulting from differences in the refractive indices of water (1.33), enamel (1.62) and air (1.0).[54] A white spot lesion is a porous surface, and the pores are filled with saliva when the lesion is wet. Because the refractive indices of water and enamel are so close, an early white spot lesion can hardly be distinguished from a sound enamel surface in the presence of saliva. When a white spot lesion is thoroughly dried, air replaces the saliva in the pores, making the lesion frostier and more clearly seen.

  Furthermore, drying the tooth surface during examination has diagnostic as well as prognostic functions. Removing water from the porous tissue enables the dentist to gauge how far through the enamel a lesion has progressed. A white spot lesion visible on a wet tooth surface indicates that demineralization is over halfway through the enamel, possibly extending into dentin. A white spot lesion that becomes visible only after thorough air drying will be less than halfway through enamel.[55–58] Prognostically, the latter condition has a better chance of complete reversal, while the former condition may only be arrested.[57]
- *Magnification and lighting.* It has been demonstrated that the accuracy of visual caries detection can be improved 50% with the use of a magnifying device, such as a prism loupe or a surgical microscope.[59] Good lighting aids vision. The mouth is a dark hole and the operating lamp illuminates it. The dental mirror can catch and reflect the light onto individual tooth surfaces.

General clinical criteria for visual caries detection. The identification of a caries lesion is a stepwise process. The diagnosis of a caries lesion should be preceded by

a series of questions so that the most appropriate type of intervention can be selected for the patient (ie, it provides a basis for a treatment decision)52: Is there a change from normal appearance of enamel or root surface (presence of a lesion)? Is the change consistent with the appearance of dental caries or developmental hypomineralization? If there is caries, is it cavitated (loss of surface integrity) or noncavitated? If it is noncavitated, is there a shadowing underneath the enamel or white (or brown/black) spot lesion? If there is a white (or brown/black) spot lesion, is it active or arrested?

The whitish appearance of a lesion is due to changes in the optical properties of enamel with demineralization or hypomineralization. Hence, carious white spots should be differentiated from developmental hypomineralizations such as fluorosis. The following factors may guide the dentist to a definitive diagnosis of a white spot as caries or developmental hypomineralization (Table 5-4).

Table 5-4 **Differences between white spots due to caries or developmental hypomineralization**

Characteristic	Caries	Hypomineralization
Appearance and texture	Opaque, chalky, and matte (rough) surface when air dried	Smooth, glossy, and less opaque surface when air dried
	May be discolored by extrinsic stain	No extrinsic discolorat
	Feels rough when the tip of a CPITN probe or blunt explorer is moved gently across the surface	Feels smooth when the tip of a CPITN probe or blunt explorer is moved gently across the surface
Location and distribution	Located in plaque stagnation areas	Located mainly in self-cleansing areas
		Bilateral or multiple corresponding teeth; eg, canines

CPITN–Community Periodontal Index of Treatment Needs.

Appearance and texture. After prolonged drying (> 5 seconds) with an air-water syringe, white spot lesions from developmental hypomineralization usually have smooth and glossy surfaces like a sound enamel surface, while active caries lesions have chalky, matte (rough) surfaces. This can be elicited by gently running the tip of a ball-ended Community Periodontal Index of Treatment Needs (CPITN) probe or blunt explorer on the surface of the lesion. Carious white spots can be discolored

after a long exposure in the oral environment, while the surface of developmental hypomineralization is not porous and cannot be easily discolored, except in mottling due to severe fluorosis.

Location and distribution. Caries occurs in plaque stagnation areas (areas that are difficult to reach by toothbrush), such as gingival margins, occlusal pits and fissures, buccal/lingual grooves and fossa, proximal areas below contact points, and stagnant areas created by overcrowding of teeth. Although developmental hypomineralization can occur at any point on the tooth surface, a white spot at the tip of the cusp or coronal third of a tooth is more likely to be a developmental anomaly because these areas are self-cleansing and harbor hardly any plaque. Moreover, white spots that occur on corresponding sites and have similar shape on all corresponding teeth (eg, all first premolars) in the four quadrants of the dentition are likely to be developmental hypomineralizations. These sites on those four teeth developed at the same time and would be equally affected by tooth mineralization factors.

Clinical visual caries scoring system. Efforts to improve examiner objectivity and caries status recording has led to the development of scoring systems to stage lesions along the continuum—from very small initial lesions, just visible to the human eye, through established white and brown spot lesions, to shadowing beneath the enamel, and to different extents of cavitation.[53] The International Caries Detection and Assessment System (ICDAS) is highly recognized among these systems.[60] It was developed for use with primary coronal caries, caries adjacent to restorations and sealants, and for root surface caries and includes noncavitated caries lesions. The ICDAS detection codes (Table 5-5) include a seven-point staging of caries, which enable a detailed recording of caries status during a patient's initial assessment comparison over time. It is also applied in caries risk evaluation and reevaluation. ICDAS is a two-stage, two-digit coding system comprised of a code for the restoration and sealant status of the tooth surface, followed by a code for the caries severity (see Table 5-5).

Table 5-5	**ICDAS criteria for visual examination of caries lesions, including two-digit coding for restoration and caries severity[53] The International Caries Detection and Assessment System (ICDAS) is highly recognized among these systems.[53,53] The International Caries Detection and Assessment System (ICDAS) is highly recognized among these systems.[60]**
First digit (restoration and sealant	

codes)	Second digit (caries severity codes)
0 = Not sealed or restored	0 = Sound tooth surface
1 = Partial sealant	1 = First visual change (opacity or discoloration) in enamel hardly visible on the wet surface but distinctly visible after air drying
2 = Full sealant	2 = Distinct visual change (opacity or discoloration) in enamel, visible without air drying
3 = Tooth-colored restoration	3 = Enamel breakdown, no dentin visible
4 = Amalgam restoration	4 = Dentin shadow (not cavitated into dentin)
5 = Stainless steel crown	5 = Distinct cavity with visible dentin
6 = Porcelain, gold, PFM crown or veneer	6 = Extensive distinct cavity with visible dentin
7 = Defective restoration	
8 = Temporary restoration	

PFM–porcelain fused to metal.

Although the ICDAS caries severity codes 1 and 2 of noncavitated caries (white or brown spot lesion) are based on clinical appearance of the lesion in relation to moisture on the surface, it has value for diagnosis, treatment, and prognosis. For occlusal surface caries, there is a high correlation between the ICDAS caries severity codes and the histologic depth of the lesions into enamel and dentin[55–58] (Table 5-6). Studies have demonstrated histologically that in ICDAS code-1 lesions the enamel demineralization is limited to the outer half of the enamel layer[55–58]; this lesion has higher chances of complete reversal with remineralization therapy. In an ICDAS code-2 lesion, the demineralization involves the entire enamel layer and outer third of the dentin layer,[55–58] and this lesion stands higher chances of being arrested rather than reversed.

Table 5-6	Correlation of the ICDAS caries severity codes with the histologic depth of the lesions into the tooth tissue on the occlusal surface,[55–58]
Score	Demineralization depth
0	No enamel demineralization

1	Enamel demineralization limited to the outer half of the enamel layer
2	Demineralization involving enamel and the outer third of the dentin
3-4	Demineralization involving the middle third of the dentin
5-6	Demineralization involving the pulpal third of the dentin

The ICDAS Committee also developed an International Caries Classification System (ICCS), a simplified format of ICDAS codes for general practice[2] (Table 5-7). This format groups the ICDAS codes into four color-coded stages of caries.[2] The color coding facilitates communication between patients and the dental team.

Table 5-7 **ICCS codes for general practice***

Sites of caries (1–4)	Stages of caries (0–3)			
	0 No disease ICDAS 0	1 Initial lesion ICDAS 1 and 2	2 Moderate lesion ICDAS 3 and 4	3 Extensive lesion ICDAS 5 and 6
1. Pits and fissures	1.0	1.1	1.2	1.3
2. Approximal surfaces	2.0	2.1	2.2	2.3
3. Cervical and smooth surfaces	3.0	3.1	3.2	3.3
4. Root surfaces	4.0	4.1	4.2	4.3

*Reprinted from Pitts and Richards[2] with permission.
0 *(green)*–sound tooth surface; 1 *(yellow)*–distinct, visual noncavitated change in enamel; 2 *(orange)*–localized enamel breakdown; 3 (red)—distinct cavity with visible dentin.

Visual caries detection specific to different types of lesions. Pit and fissure lesions. Detection of carious structure in fissures and pits is most often performed by visual inspection, and good lighting and dry, clean teeth are prerequisites. Any sign of visible cavitation in the occlusal surface corresponds to progression of the lesion into the dentin.[55,56,61] When occlusal surface caries was recorded visually using a caries score ranking system (see Table 5-5), a high correlation with the histologic depth of the lesions into enamel and dentin was found[55,56,61] (see Table 5-6).

Shadowing or grayish discoloration of the adjacent translucent enamel along the pits and fissures may indicate possible undermining of enamel by hidden caries. Opaque, matte texture of enamel adjacent to the stained pits and fissures may indicate the presence of active caries underneath them. When there is no caries underneath, discoloration will remain within the confines of the pit and/or fissure. However, lesions underneath (along the walls) will be discolored by stain that diffused through the porous lesion surface. Looking perpendicularly through the translucent adjacent healthy enamel along the pit and/or fissure, a clinician can observe these stained "wall" lesions as discoloration extending beyond the confines of the pit and/or fissure in the form of a "bottle-brush" appearance (Fig 5-10). These lesions are caused by plaque stagnation at the entrance to the fissure. When this plaque is cleaned off, the lesion is clearly visible. The patient can arrest this lesion by brushing with toothpaste containing fluoride.[62]

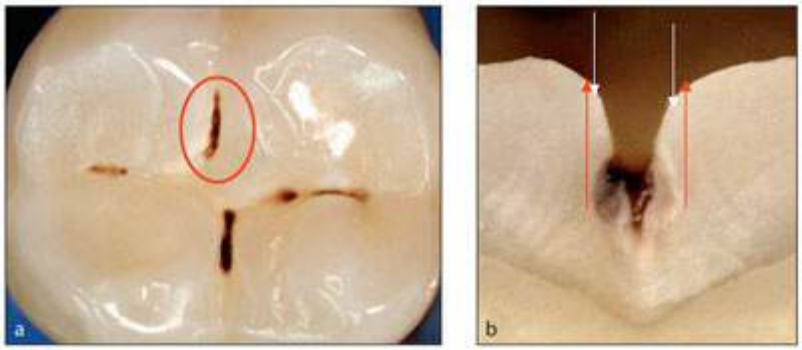


Fig 5-10 Illustration of the visual detection of lesions on the walls of stained pits/fissures. *(a)* A stained fissure *(red circle)* under visual examination. *(b)* Looking perpendicularly through the translucent adjacent healthy enamel along the fissure *(white arrows)*, lesions on the walls of the fissure are seen as discoloration extending (ie, underneath the translucent enamel) beyond the confines of the fissure with a "bottle-brush" appearance *(red arrows)*.

Careful examination of bitewing radiographs is also important, although occlusal enamel lesions will not be visible. Carious dentin, however, can usually be detected, and such lesions are often large (Fig 5-11). Bitewing radiographs provide a "safety net" for detecting occlusal lesions.[57,63]



Fig 5-11 The radiograph as "safety net": Occlusal carious dentin is well detected (man-dibular left first molar).

Tactile examination of fissures with a dental explorer has long been advocated as an important method to detect caries lesions, but research has shown this to be an unwise practice. The method is inaccurate,[64,65] and, worse, the explorer can damage a white spot lesion by breaking through the relatively intact surface zone[66] (Fig 5-12). Vigorous use of a sharp explorer can cause a cavity that will subsequently trap dental plaque and encourage lesion progression. Detection of fissure caries lesions should rely on sharp eyes, not sharp explorers. The clinician should use the explorer in the following appropriate ways:

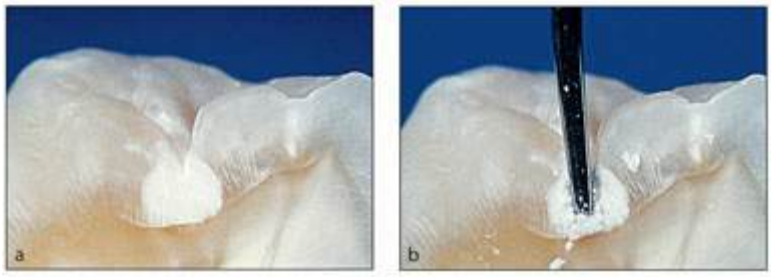


Fig 5-12*(a and b)* The explorer tip can easily damage white spot lesions.

- Detect and remove plaque in fissures, along gingival margins, and in proximal spaces.
- Gently "feel" margins and defects to confirm and assess cavitations. • Gently "feel" the hardness of root surface lesions.
- Gently "feel" the texture (roughness) of white spot lesions by gentle running of the tip of a ball-ended CPITN probe or blunt explorer on the surface of the lesion.

Lesions involving proximal surfaces. Extensive active proximal lesions can be revealed by shadowing or grayish discoloration of the undermined occlusal enamel ridge. Otherwise, when there is contact between proximal surfaces, the radiograph is the most accurate method for detecting demineralization. In the premolar and molar regions, lesion progression or arrest can be monitored, provided that subsequent bitewing radiographs are taken at approximately the same angulation.[66] The radiograph should be examined carefully to determine whether caries lesions are present in the outer enamel, at the dentinoenamel junction, in the outer half of dentin, or in the inner half of dentin (Fig 5-13).

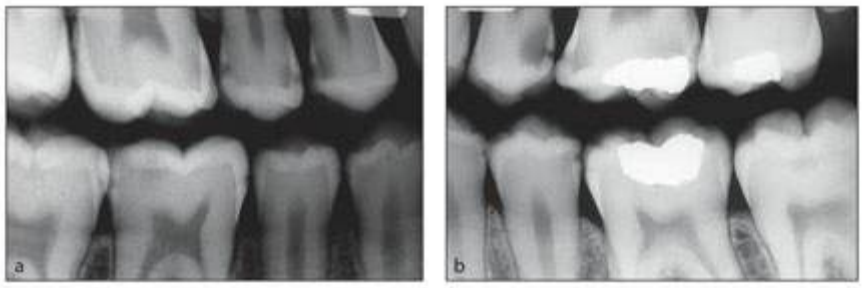


Fig 5-13*(a and b)* Radiographs showing proximal demineralizations in the outer enamel to the dentinoenamel junction and in the outer and inner half of dentin. (Occlusal lesions are visible on the mandibular left second molar and right first and second molars, and a recurrent caries lesion is underneath the restoration on the maxillary left first molar.)

For detection of proximal lesions in anterior teeth, the fiber-optic transillumination technique is particularly appropriate and convenient.[67] With this technique, a fine light, coned down to a 0.5-mm diameter, is transmitted through a contact area. A lesion appears as a dark shadow. It is difficult, however, to discriminate between demineralization extending just into enamel and that progressing further into dentin, especially in the posterior areas.

Finally, use of an orthodontic separator has been advocated in some cases to allow the dentist to see more clearly and to gently feel for a break in the enamel surface.[68]

Lesions on buccal, lingual, and root surfaces. Active caries lesions in smooth, free enamel surfaces are typically located close to the gingival margin and have chalky matte, whitish/ yellowish surfaces. A caries lesion on a root surface is seen as a clearly demarcated, light brown, dark brown, or black discolored area on the root surface or at the cementoenamel junction.[69] A root surface lesion is *cavitated* if there is loss of surface integrity or if the depth of the cavity is . 0.5 mm when measured with the ball of the CPITN probe. The diameter of the ball is 0.5 mm, so if the cavity accommodates the ball, then the depth is . 0.5 mm. If there is no loss of surface integrity or if the depth of the cavity is less than 0.5 mm (does not accommodate the ball), the lesion is *noncavitated*.[69]

Secondary or recurrent caries. *Secondary caries* o r *recurrent caries* are primary lesions at the margins of a restoration. The clinical criteria are thus identical to those for primary caries. The dentist can check that the restoration is secure in the cavity by pushing it with an explorer; a loose restoration ensures plaque stagnation, and active caries is likely to be present.

Radiographic examination

Bitewing radiography has a long history in the dentist's armamentarium for the detection of lesions that are clinically hidden from a careful clinical visual examination (eg, proximal caries). The importance of radiographs to diagnose initial caries lesions in proximal surfaces is well established.[70] It is possible to detect early proximal lesions amenable to preventive care. However, radiographic evaluation of occlusal surfaces has been found to be of minimal diagnostic value for detecting enamel caries and superficial dentinal caries because of the large amounts of surrounding sound enamel. In deeper occlusal lesions reaching the dentin, radiographs show a high *specificity* (the identification of true negatives). Although useful for proximal surfaces, the bitewing radiograph still cannot distinguish between cavitated and noncavitated proximal lesions; thus, it is necessary to establish what level of radiolucency should be accepted as cavitation.[71] This threshold will aid in determining the treatment strategy to be applied for a particular lesion (Fig 5-14). The threshold determines the *sensitivity* (the identification of true positives) and specificity of detection decisions. A low diagnostic threshold leads to detection of not only more small lesions but also more false-positive diagnoses, while a high threshold will increase the specificity but will lower the sensitivity because the small lesions will be missed. Although bitewing radiographs can be used to estimate the depth of the lesion, the radiographic extent of radiolucency is not precisely correlated to the histologic extent of the lesion. Radiographs have been shown to underestimate the depth of the lesion.[72] Bitewing radiography is associated with the problem of variation in the quality of radiographs and differences in the viewing conditions. Radiographs should be obtained using a film holder and beam-aiming device to take the guesswork out of tube alignment and allow comparable views to be taken on subsequent occasions.[66] Because lesions confined to enamel should be managed by preventive treatment, radiographic monitoring is important.

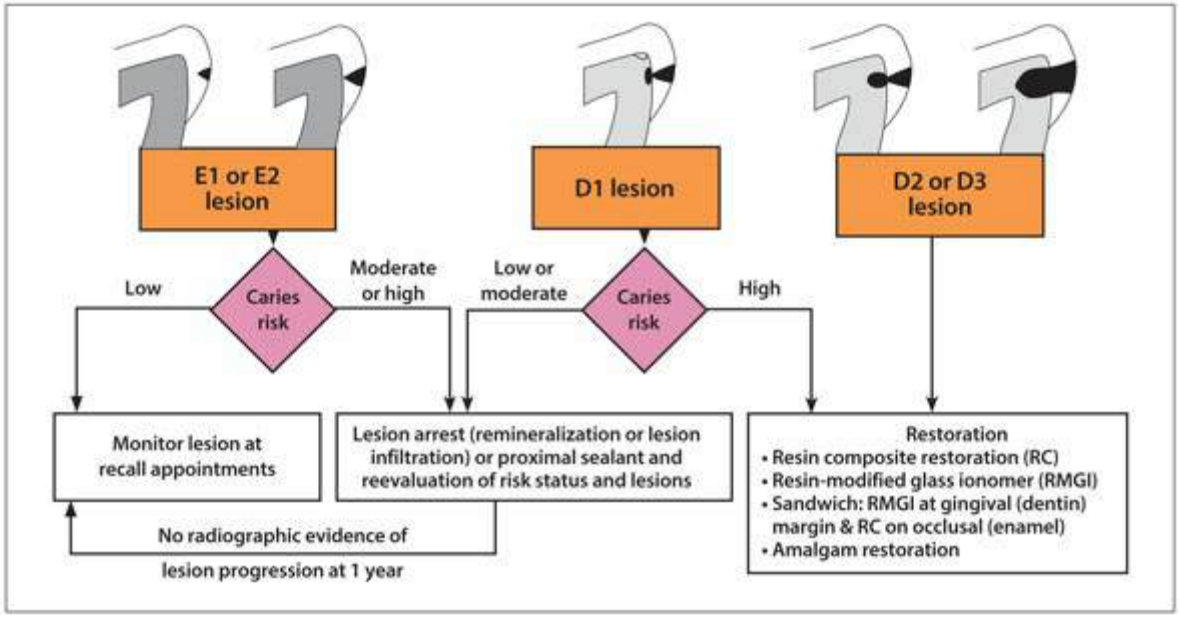


Fig 5-14 Algorithm for diagnosis and management of caries lesions in proximal coronal surfaces of permanent teeth based on radiographic classification of lesions. (Courtesy of the Department of Comprehensive Dentistry, University of Texas Health Science Center at San Antonio.) E1—less than halfway through proximal enamel; E2—more than halfway through proximal enamel, not penetrating past the dentinoenamel junction; D1—slightly past the DEJ; D2—less that halfway through dentin toward the pulp; D3—halfway or more through dentin.

Computer enhancement tools have been developed to allow images from video or digital radiographs to be enlarged to aid in early diagnosis. In addition, color enhancement increases visual contrast and has been proven to be an objective and reproducible methodology.[73–76] The Logicon Caries Detector software (Northrup Grumman)[73–75] and MiPACS Dental Enterprise Viewer (Medicor Imaging) are currently being used in conjunction with digital radiographs for clinical caries detection. The use of digital radiography permits image manipulation and reduces the amount of radiation required to obtain a diagnostic image. Although processing of digitized and directly captured images tended to perform more accurately than conventional radiography, studies have shown no significant difference between the two methods.[76] Thus, digital intraoral radiographs appear to be as accurate as film radiographs for the detection of dental caries. The sensitivity of digital systems is relatively high for detection of occlusal lesions into dentin but is of no value for detection of early enamel lesions.[76] Just like the conventional radiographic films, the main disadvantage of the digital radiography is the high number of false-positive results when used for occlusal caries detection.[77]

Caries detection aids

Caries is a dynamic process of alternating demineralization and remineralization that may result in progression, regression (reversal), or arrest (stabilization) of the lesion. Thus, the activities of a caries lesion continue to change over time as a result of continued change in mineral status of the lesion. Hence, there is a need for devices that aid in monitoring by quantitatively assessing the mineral changes over time within a caries lesion. The conventional caries detection methods, visual and radiographic examination, cannot quantitatively account for this dynamic of caries.

Diagnostic techniques capable of providing valid, reliable, and nonsubjective measurements are in demand. The readings in an acceptable instrument must be consistent between different models for the same operator on different occasions and for different operators. There are an increasing number of new technologies being developed to aid detection and monitoring of caries lesions. Most of these devices claim to detect and monitor caries activities based on the change in the reflectance, transmission, fluorescence, electrical conductance/ impedance, or ultrasound transmittal properties of enamel following demineralization. These claims have not been evaluated using controlled clinical trials. Although such instruments cannot make the clinical judgment of lesion activity when used once, a valid and reproducible method could be used longitudinally to measure lesion progression or arrest.

Diagnosis

Clinical observations have demonstrated that caries lesions can be arrested at any stage of lesion development provided that clinically plaque-free conditions are maintained. Thus, a caries lesion detected in a clinical examination can be active or inactive. An active lesion is considered to have a greater likelihood of transition (progress, arrest, or regress) than an inactive lesion. Hence, it must be determined whether the detected lesion is active or inactive. If the lesion is arrested, no treatment is required except for esthetic or functional reasons. If the lesion is active, treatment is needed to arrest lesion progression and must include preventive measures. It may also include restorative dentistry. Thus, caries diagnosis includes both caries detection and the assessment of lesion activity.[78]

Assessing caries activity

There are some features of individual lesions that indicate whether a lesion is active or arrested. These assessments are inevitably “best guesses,” but this is acceptable because most patients see their dentist at regular intervals. Thus, the initial diagnosis

should be refined at recall visits, and the effects of preventive treatment can be assessed. Table 5-8 lists parameters relevant in the assessment of caries activity.

Table 5-8	Characteristics of active and arrested lesions in enamel and roots		
Condition	**Tooth tissue**	**Active**	**Arrested**
Appearance	Enamel	Lesion is chalky (lacks luster) when air dried and is whitish or yellowish	Lesion has shiny surface when air dried and may be whitish, brownish, or blackish
	Root	Clearly demarcated and may be discolored yellowish/light brown *Note: Color change is not a reliable indicator*	Same as active lesion
Texture	Enamel	Feels rough when the tip of the probe (explorer) is moved gently across the surface	Feels hard and more smooth than rough when the tip of the probe is moved gently across the surface
	Root	Lesion has soft or leathery base on gentle probing using blunt or CPITN probe	Lesion has hard and shiny base using blunt or CPITN probe
Translucency	Enamel	Opaque	Semitransparent
	Root	NA	NA
Presence of plaque	Enamel	Lesion covered by plaque	Generally plaque free
	Root	Lesion covered by plaqu	Generally plaque free
Locations	Enamel	Mostly on plaque stagnation areas	Lesion area is usually plaque free
	Root	Anywhere	Same as active lesi
Cavitation	Enamel	From localized surface defect (microcavity) in enamel only to distinct cavity with soft or leathery base on gentle probing	From localized surface defect (microcavity) in enamel only to distinct cavity with hard base on probing with gentle pressure; surface of cavity may be shiny
		Cavitated if there is loss of surface integrity or if the depth is ≥ 0.5 mm measured	

	Root	using the ball of the CPITN probe; noncavitated if no loss of surface integrity or if the depth is < 0.5 mm	Same as active lesion

NA–not applicable.

The features of active and arrested lesions[79] at the individual sites are discussed in the following section.

Occlusal lesions. The visual features of occlusal caries lesions were presented in Table 5-5. The occlusal surface must be cleaned by the dentist with a brush before it is examined.

The following features indicate lesion activity:

- White spot lesions that have a matte or visibly frosted surface or are plaque covered (visible after drying or application of a disclosing solution)[62]
- Cavitated lesions, including small cavities and cavities exposing dentin (Fig 5-15)



Fig 5-15 Active occlusal caries.

- Lesions visible in dentin on bitewing radiographs[80]

The lesion may be arrested if white or brown spot lesions with a shiny surface[62] are present (Fig 5-16).

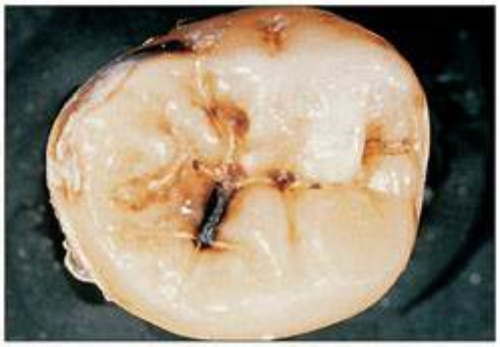


Fig 5-16 Arrested lesions.

Proximal lesions. Diagnosis of lesion activity is more difficult when the adjacent tooth precludes a direct visual assessment. (The radiographic features of proximal lesions are presented in chapter 2 and in preceding paragraphs.) The presence or absence of a cavity is relevant to lesion activity, but this cannot be determined from the radiograph unless the lesion is very deep. For an active caries lesion to be arrested, plaque must be removed regularly from the cavity. However, on a proximal surface, there is no access for the toothbrush, and even the most fastidious of flossers will only slide over the surface.

The following features tend to indicate lesion activity:

- A patient with proximal lesions on the radiograph who is at high risk for caries[71]
- A proximal lesion present radiographically plus persistent gingival inflammation despite the patient's attempts to remove plaque with dental floss[81]
- A lesion not present at previous examination

The following features indicate that the lesion may be arrested:

- Successive, reproducible bitewing radiographs showing no lesion progression
- A patient who is now judged to be at low risk for caries because, following preventive behavior, he or she presents with no new lesions

Smooth, free surface lesions. These are probably the most straightforward lesions for assessment because they are the most visible. Of all lesions, these are the ones most likely to be arrested by preventive treatment alone.

The following features indicate lesion activity:

- White spot lesions close to the gingival margin that have a matte or visibly frosted surface; these are often plaque covered (Fig 5-17)



Fig 5-17 Matte, white, active cervical lesions.

- Cavitated, plaque-covered lesions with or without exposed dentin; if dentin is exposed and soft, the dentin is heavily infected and the lesion is active (Fig 5-18)



Fig 5-18*(a and b)* Cavitated, active cervical lesions.

The following features indicate that the lesion is arrested:

- Shiny white or brown lesions, often well exposed due to gingival recession; the lesions are not plaque covered (Fig 5-19)



Fig 5-19 Shiny, white, arrested cervical lesion.

- Cavitated lesions, often dark brown, with hard dentin at their bases; the lesions are not plaque covered and are often remote from the gingival margin (Fig 5-20)



Fig 5-20 Brown, arrested lesion.

Root caries. Active lesions are:

- Close to the gingival margin and plaque covered
- Soft or leathery in consistency

Arrested lesions are:

- Often at some distance from the gingival margin and not covered with plaque (Fig

5-21)



Fig 5-21 Arrested root caries lesions.

• As hard as the surrounding healthy root surface[82]

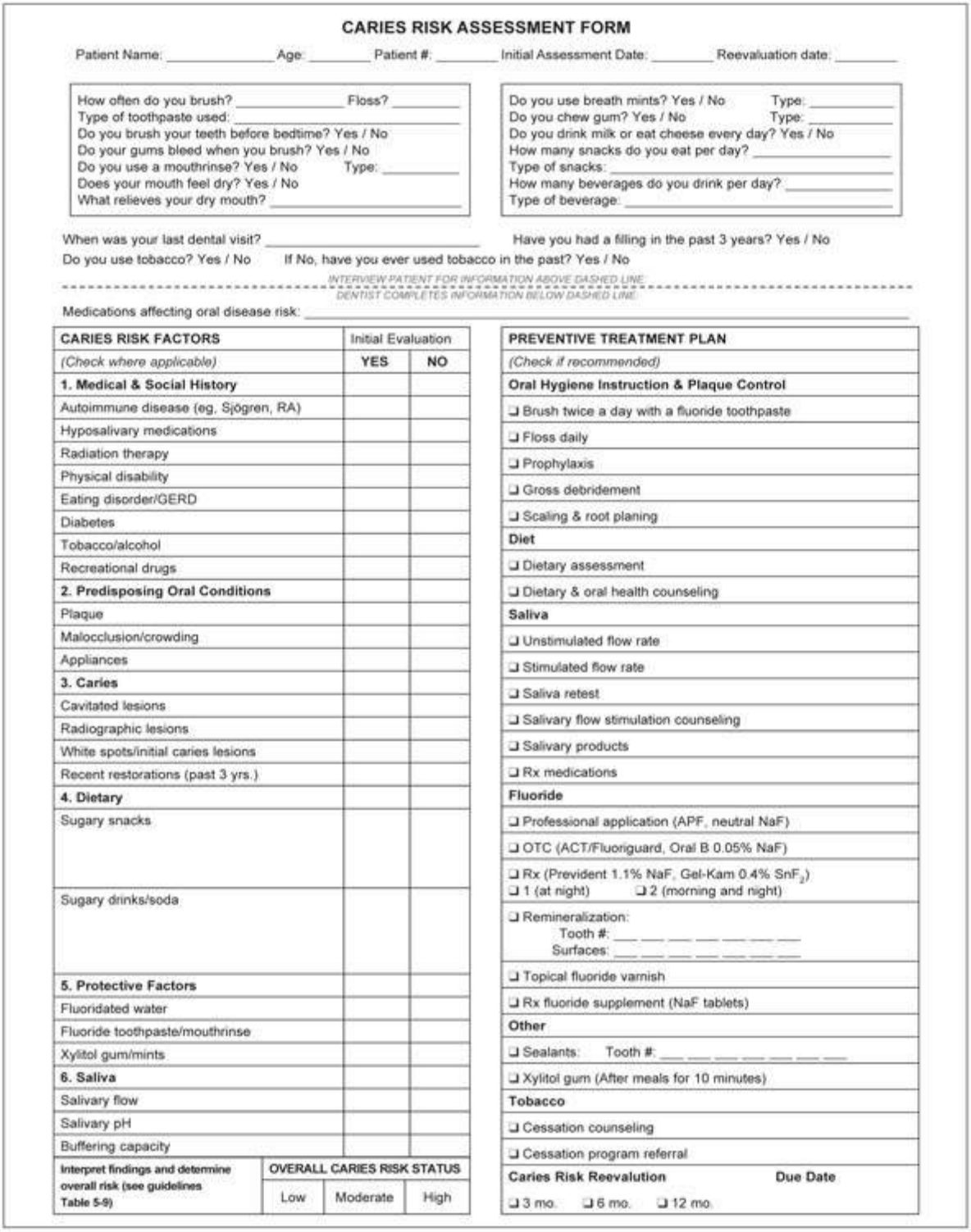
Color is unreliable in differentiating active from inactive lesions. While arrested lesions may be dark, so may be the soft dentin in some active lesions. There currently is no clear explanation of why demineralized dentin is brown.

Recurrent caries. A caries lesion at the margin of a restoration is called *recurrent caries,* and its diagnosis on various tooth surfaces is the same as that described previously for primary lesions. The bitewing radiograph is very important in the diagnosis of recurrent caries lesions because they often form cervical to an existing proximal restoration, in the area of plaque stagnation. Lesions that are obviously in dentin as seen radiographically tend to be cavitated and active. However, research has shown that ditching and staining around an amalgam restoration[83] and staining around a tooth-colored restoration[84] are not reliable indicators of an active caries lesion beneath a restoration.

Assessing Caries Risk

Caries risk may be defined as the probability of an individual developing a certain number of caries lesions reaching a given stage of disease progression during a specified period.[85] It is a key and essential component of PCCM.[2] Because caries is a multifactorial disease, it is not surprising that the format of existing tools for caries risk assessment varies among developers and depends on the information the tool is tailored to collect. However, a tool for comprehensive caries risk assessment should consist of three major sections (Fig 5-22). The first part is the *subjective section* that collects the information given by the patient on his or her *(1)* oral hygiene attitude, including dental plaque control and use of fluoride/other anticaries agents; *(2)* dietary habits; *(3)* access to dental care; *(4)* use of tobacco; and *(5)* use of medications affecting caries risk. The second part is the *objective section* that

contains information gathered about the patient by the dentist through clinical examination, clinical tests (if applicable), and conversation and is composed of the *(1)* medical history, *(2)* social history, *(3)* existing caries and restorations, *(4)* saliva properties, and other oral conditions predisposing the patient to caries risk. The third part is the *synthesis section* that identifies the overall risk level (low, moderate, high) of the patient. Some tools may contain a fourth section that lists the preventive care options from which the clinician can develop a personalized preventive plan for an individual patient.



CARIES RISK ASSESSMENT FORM

Patient Name: _____ Age: _____ Patient #: _____ Initial Assessment Date: _____ Reevaluation date: _____

How often do you brush? _____ Floss? _____
Type of toothpaste used: _____
Do you brush your teeth before bedtime? Yes / No
Do your gums bleed when you brush? Yes / No
Do you use a mouthrinse? Yes / No Type: _____
Does your mouth feel dry? Yes / No
What relieves your dry mouth? _____

Do you use breath mints? Yes / No Type: _____
Do you chew gum? Yes / No Type: _____
Do you drink milk or eat cheese every day? Yes / No
How many snacks do you eat per day? _____
Type of snacks: _____
How many beverages do you drink per day? _____
Type of beverage: _____

When was your last dental visit? _____ Have you had a filling in the past 3 years? Yes / No
Do you use tobacco? Yes / No If No, have you ever used tobacco in the past? Yes / No

INTERVIEW PATIENT FOR INFORMATION ABOVE DASHED LINE
DENTIST COMPLETES INFORMATION BELOW DASHED LINE

Medications affecting oral disease risk: _____

CARIES RISK FACTORS	Initial Evaluation	
(Check where applicable)	YES	NO
1. Medical & Social History		
Autoimmune disease (eg, Sjögren, RA)		
Hyposalivary medications		
Radiation therapy		
Physical disability		
Eating disorder/GERD		
Diabetes		
Tobacco/alcohol		
Recreational drugs		
2. Predisposing Oral Conditions		
Plaque		
Malocclusion/crowding		
Appliances		
3. Caries		
Cavitated lesions		
Radiographic lesions		
White spots/initial caries lesions		
Recent restorations (past 3 yrs.)		
4. Dietary		
Sugary snacks		
Sugary drinks/soda		
5. Protective Factors		
Fluoridated water		
Fluoride toothpaste/mouthrinse		
Xylitol gum/mints		
6. Saliva		
Salivary flow		
Salivary pH		
Buffering capacity		

Interpret findings and determine overall risk (see guidelines Table 5-9)	OVERALL CARIES RISK STATUS		
	Low	Moderate	High

PREVENTIVE TREATMENT PLAN
(Check if recommended)
Oral Hygiene Instruction & Plaque Control
❑ Brush twice a day with a fluoride toothpaste
❑ Floss daily
❑ Prophylaxis
❑ Gross debridement
❑ Scaling & root planing
Diet
❑ Dietary assessment
❑ Dietary & oral health counseling
Saliva
❑ Unstimulated flow rate
❑ Stimulated flow rate
❑ Saliva retest
❑ Salivary flow stimulation counseling
❑ Salivary products
❑ Rx medications
Fluoride
❑ Professional application (APF, neutral NaF)
❑ OTC (ACT/Fluoriguard, Oral B 0.05% NaF)
❑ Rx (Prevident 1.1% NaF, Gel-Kam 0.4% SnF_2) ❑ 1 (at night) ❑ 2 (morning and night)
❑ Remineralization: Tooth #: ___ ___ ___ ___ ___ ___ ___ ___ Surfaces: ___ ___ ___ ___ ___ ___ ___ ___
❑ Topical fluoride varnish
❑ Rx fluoride supplement (NaF tablets)
Other
❑ Sealants: Tooth #: ___ ___ ___ ___ ___ ___ ___ ___
❑ Xylitol gum (After meals for 10 minutes)
Tobacco
❑ Cessation counseling
❑ Cessation program referral
Caries Risk Reevalution **Due Date** ❑ 3 mo. ❑ 6 mo. ❑ 12 mo.

Fig 5-22 Caries risk assessment form. (Modified from the Department of Comprehensive Dentistry, University of Texas Health Science Center at San Antonio with permission.)

Identifying the risk factors

Presence of caries

The presence of cavitated, noncavitated (initial lesions), and/ or radiographic (mainly proximal lesions) caries lesions is an indication that the patient is actively developing new caries that are progressing from initial caries to visible cavitations. Any of these findings is an indication that the patient is at risk of developing caries, and they should be carefully assessed, along with other caries risk factors, to determine the patient's overall caries risk.

Recent restorations (one or more within the past 3 years) is also an indication that the patient has been active in developing caries and should be seen to be at a risk of developing further caries, which places the patient in a high-caries-risk category. Combining this information with the date of the last dental visit gives an indication of the patient's caries activity over a period of time and also indicates the ability and interest of the patient in seeking dental care.

Dental plaque

Dental plaque is a risk factor for dental caries.[86] All patients with active lesions should be shown the relationship of disclosed plaque in the mouth to the lesions that are present. This makes advice on tooth cleaning relevant to the patient and specific to the lesion.

Note should be taken of the following experimental evidence[87]:

- Fluoride concentration of toothpaste. Confirm that the patient is using a toothpaste containing fluoride (1,000 ppm and above). Although the international recommendation is 1,000 ppm fluoride for younger children and up to 1,500 ppm for older children, the decision of what fluoride levels to use for children under 6 years should be balanced with the risk of fluorosis. The possible risk of fluorosis may necessitate recommending using toothpaste containing less than 1,000 ppm fluoride.[88]
- Frequency of brushing. Fluoride-containing toothpaste should be used twice a day, including immediately before bedtime.
- Toothbrushing in children. Check that a parent or caregiver is helping a child to brush and supervising the amount of toothpaste on the brush for children younger than 7 years of age. The adult should complete the brushing, paying particular attention to occlusal surfaces of erupting teeth.
- Rinsing behavior. Ask whether large volumes of water are used to wash away excess toothpaste. The message should be to "spit, don't rinse" or to use the

toothpaste slurry as rinse before spitting. This maintains the fluoride in contact with the teeth longer.

- Ability to remove plaque. When plaque is disclosed, can the patient remove it? On a subsequent visit, has he or she removed plaque? The answer to the second question defines the patient's motivation rather than his or her ability with a toothbrush.

It should also be remembered that placing a dental appliance such as an orthodontic appliance[89] or a partial denture[90] in the mouth can encourage plaque retention and explain the development of caries lesions.

Use of fluoride

Check the following:

- Does the patient's toothpaste contain fluoride?
- Does the patient live in an area in which the water is fluoridated?

The patient with multiple lesions is unlikely to live in an area with fluoridated water. It is also possible that he or she has selected (perhaps unwittingly) nonfluoride toothpaste.

Diet

The diet of a patient presenting with multiple lesions, whether white spots or cavities, should be analyzed for frequency of sugar intake. An inappropriate dietary habit, such as a sweet drink frequently consumed, will often be obvious.[91]

Saliva

A dry mouth predisposes to high caries risk,[92] and measuring salivary flow, particularly unstimulated, is useful because a low flow rate may help to explain multiple lesions. There are four common causes of dry mouth:

1. Medications, such as antidepressants, antipsychotics, tranquilizers, antihypertensives, and diuretics
2. Sjögren syndrome, which affects the salivary and lacrimal glands, leading to a dry mouth and dry eyes; rheumatoid arthritis may indicate the presence of this disorder

3. Eating disorders, which may induce hyposalivation; this, combined with a poor diet, can lead to extensive caries[93]
4. Radiation therapy in the region of the salivary glands for a head and neck malignancy, which often induces xerostomia

Patients with multiple caries lesions will have high salivary counts of mutans streptococci and lactobacilli. Many research studies have shown that these counts help predict caries risk.[94] This large volume of work appears to show that, in individual patients, low counts are often good predictors of low risk but that the opposite is not necessarily true. Therefore, the routine use of salivary counts for the prediction of risk status is not recommended for everyone.

Medical history

Medically compromised and handicapped people may be at high risk for developing caries, and the onset of such problems may explain a change in a patient's caries status.[95] For these patients, oral hygiene may be difficult, and long-term use of medicines can be a problem if the medicines are sugar based[96] or cause reduced salivary flow.

The most important caries risk factor in a medical history is the complaint of dry mouth, discussed previously. It is important to realize that the medical history is a factor that can change. A vigilant dentist will detect such changes and assist the patient with the potential dental consequences.

Social history

Many studies[97,98] have shown that social deprivation can predispose to caries. The dentist may notice high caries rates in siblings, and the patient or parent may possess little knowledge of dental disease. Concern about dental health may be low and dental visits irregular.

Guide to identifying the overall caries risk status of the patient

The dentist is now in a position to categorize the caries risk status (low, moderate, high) of a patient. Identifying the caries risk status can be facilitated by categorizing the information gathered in the subjective and objective sections of the assessment tool into "risk" (pathologic) factors that may increase caries risk and "protective"

factors that may decrease caries risk[99] (Fig 5-23). Certain factors and criteria as summarized in Table 5-9 and 5-10 may help a dentist in making the decision on the overall risk level of the patient.

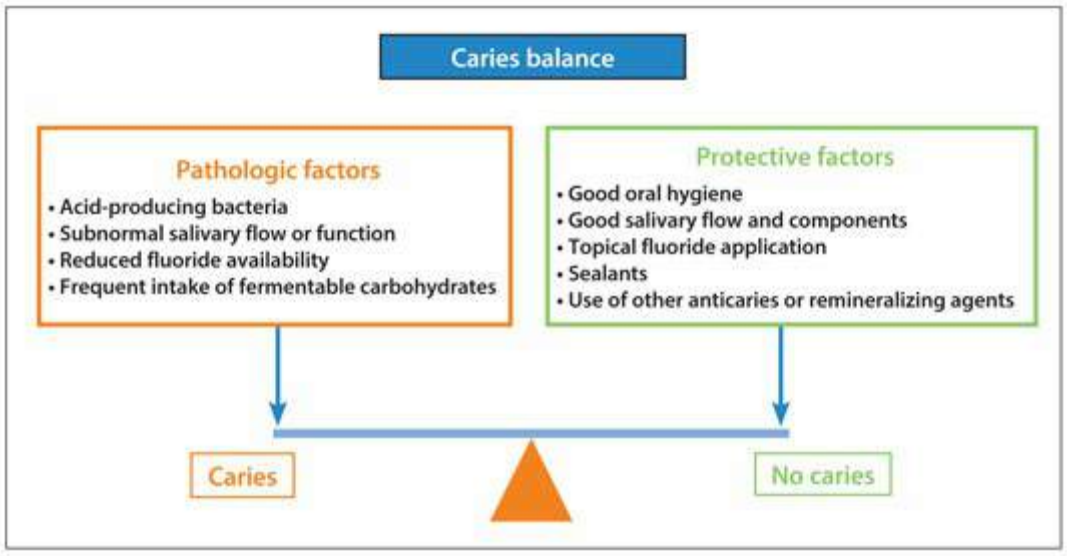


Fig 5-23 A schematic of the caries balance. (Modified from Featherstone[99] with permission.)

Table 5-9	**American Dental Association (ADA) recommended guide to identifying an overall caries risk status[100]**	
Caries risk category	**Younger than 6 years**	**6 years or older**
Low	No white spot or cavitated primary or secondary caries lesions during the last 3 years and no factors that may increase caries risk	No white spot or cavitated primary or secondary caries lesions during the last 3 years and no factors that may increase caries risk
Moderate	No white spot or cavitated primary or secondary caries lesions during the last 3 years but presence of at least one factor that may increase caries risk	One or two white spots or cavitated primary or secondary caries lesions in the last 3 years OR No white spot or cavitated primary or secondary caries lesions in the last 3 years but presence of at least one factor that may increase caries risk

High	• Any white spot or cavitated primary or secondary caries lesions during the last 3 years • Presence of multiple factors that may increase caries risk • Low socioeconomic status, especially in children too young for their risk to be based on caries history • Suboptimal fluoride exposure • Xerostomia	• Three or more white spots or cavitated primary or secondary caries lesions in the last 3 years • Presence of multiple factors that may increase caries risk • Suboptimal fluoride exposure • Xerostomia

Table 5-10	**Specific caries risk factors that may guide the overall caries risk status**	
Factor	**High risk**	**Low risk**
Plaque	Poor hygiene	Good hygiene
Type of diet	High sucrose	Low sucrose
Carbohydrate frequency	> 7 times per day	< 7 times per day
Resting SFR	< 0.2 mL/min	≥ 0.2 mL/min
Stimulated SFR	< 0.7 mL/min	≥ 0.7 mL/min
Buffer capacity	Low: ↓ pH	High: ↑ pH
Fluorides	Low: ↓ remineralization	Optimal: ↑ remineralization
Caries lesions	White spots	No white spots
Caries activity	Within last 3 years	None
Socioeconomic status	≤ Poverty level	> Twice the poverty level
Mother's caries history	High caries rate	Not caries active
Sibling caries history	High caries rate	Not caries active
Home dental care	None	Established

SFR–salivary flow rate.

Knowledge of the risk and protective factors serves both diagnostic and prognostic purposes and forms the basis for the preventive portion of the treatment plan, while the overall caries risk status forms the platform for the integrated, personalized treatment plan for the individual patient. This risk knowledge may be of great use to the dentist in explaining the lesions that are present and in identifying factors that need to be modified in order to reduce risk; poor plaque control and/or a frequently consumed sweet drink is a good example of this. Other factors that cannot

be modified may affect the preventive management regimen and the prognosis. For instance, a patient with reduced salivary flow because of Sjögren syndrome is, and will remain, at high risk for caries. Furthermore, while the dentist might like to make the prediction, "This patient will develop new caries lesions within the next 12 months unless something is done to prevent it," it is difficult to predict accurately because caries is a multifactorial disease. Clinical and radiographic examination and dental history are the most important sources of information. Thus, a person with no active lesions, few or no restorations, and no history of needing restorations replaced may safely be designated as currently at low risk. On the other hand, a person with multiple active lesions, a heavily restored mouth, and a history of repeated replacement of restorations may be designated as being at high risk. Instead of concentrating on predicting the future, dentists should concentrate on controlling the caries lesions their patients have at present. The linchpin of the preventive approach is proper self care, and this also helps to prevent future lesions.

Risk-Based Caries Management

Traditionally, radiographic evidence of demineralization in enamel or to the dentinoenamel junction has led to the immediate decision to place a restoration. Such management is still an accepted part of some state board examinations. However, contemporary research has shown that this is not the correct approach. Radiographs may reveal demineralization, but this does not necessarily mean that there is an active lesion. If the lesion is arrested, it requires no treatment. Caries is caused by a multifactorial process, and its management should reflect this.[101] The general approach to active caries should be preventive treatment,[19] including plaque control, use of fluoride, and dietary modification. Restoration makes up only one part of a strategy to facilitate plaque control. Hence, the recommended caries management strategy is PCCM, which is enabled by the ICDAS clinical visual scoring systems for caries detection and activity[53,60] and integrated into the CAMBRA treatment approach.[3,4] With PCCM, treatment should be based on interpretation of the activity of the lesion and on future caries risk. Thus, a personalized treatment plan is developed from the information gathered through the risk assessment together with the overall caries risk status of the patient. The plan should involve the choice of appropriate treatment options (preventive and noninvasive treatment options; operative and minimally invasive treatment options), followed by maintenance care, including recall visits for reassessment and monitoring.

Establishing the right protocols can help protect patients. Figures 5-24 to 5-26 (see also Fig 5-14) demonstrate the algorithms for the management of lesions in occlusal pits/fissures, nonproximal smooth surfaces, proximal surfaces, and root surfaces, with different features leading to different treatment options depending on lesion activity and risk assessment. Boxes 5-1 to 5-3 depict the general guidelines for caries management in an individual patient; these contain all the elements of personalized treatment planning and are based on the acronym *PTPM* (plaque control, treatment of existing caries lesions, protection of surfaces at risk, and maintenance for prevention).

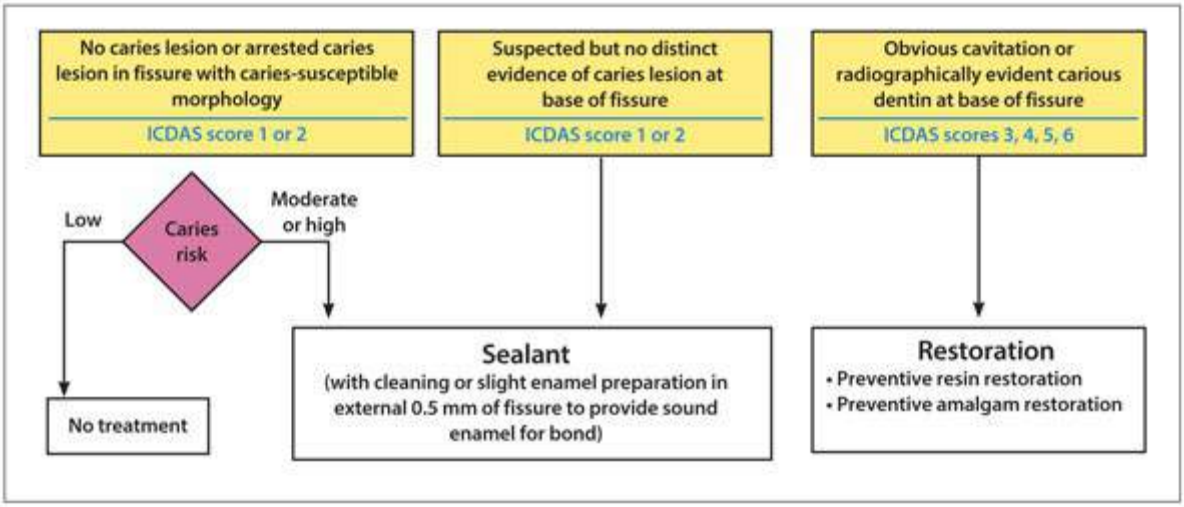


Fig 5-24 Algorithm for caries management in fissured surfaces of permanent teeth. (Courtesy of the Department of Comprehensive Dentistry, University of Texas Health Science Center at San Antonio.)

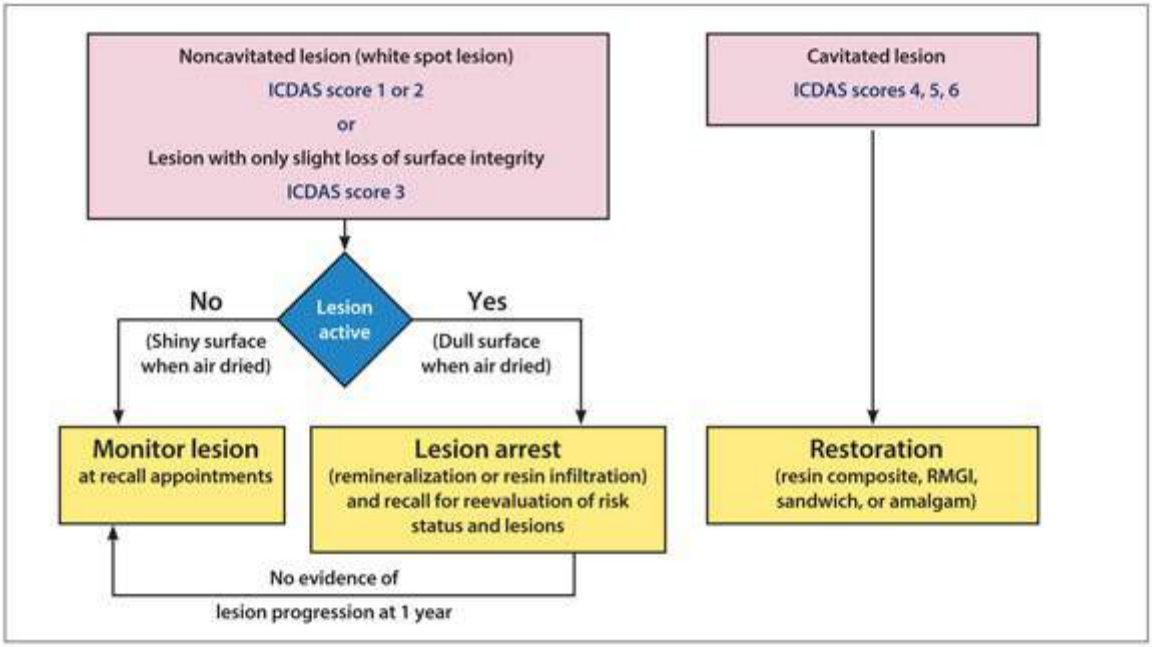


Fig 5-25 Algorithm for management of initial caries lesions in nonproximal coronal smooth surfaces of permanent teeth. RMGI—resin-modified glass ionomer. (Courtesy of the Department of Comprehensive Dentistry, University of Texas Health Science Center at San Antonio.)

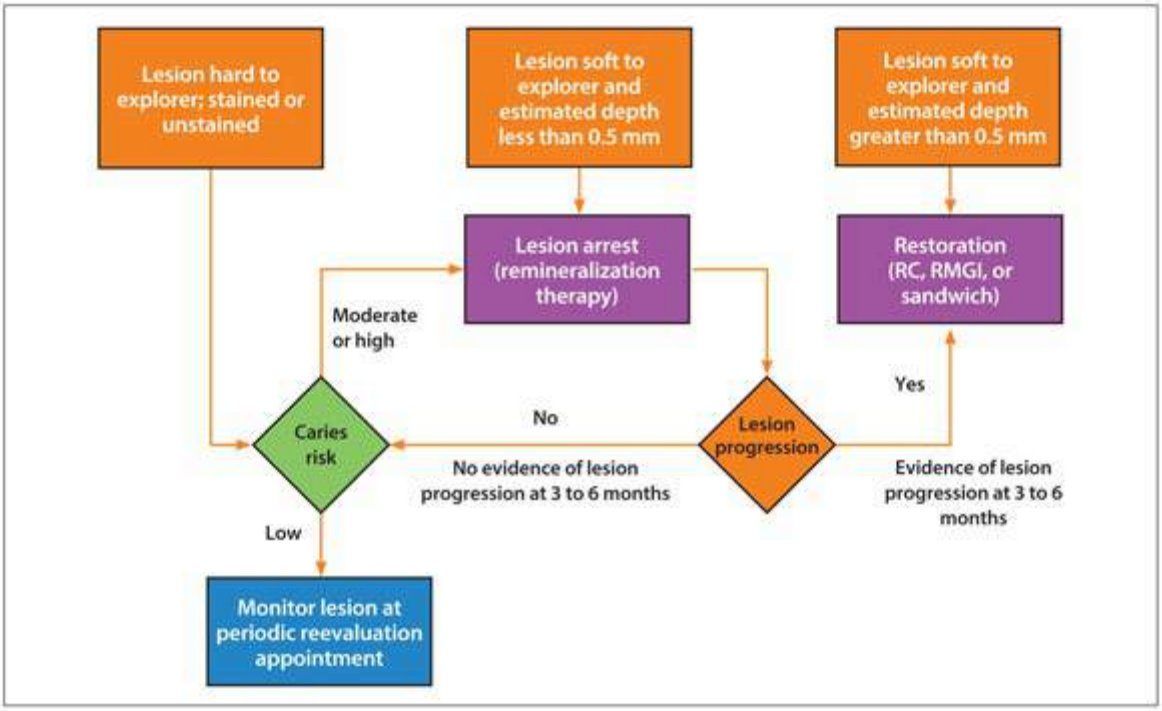


Fig 5-26 Algorithm for diagnosis and management of caries lesions in root surfaces of permanent teeth. RC—resin composite; RMGI—resin-modified glass ionomer. (Courtesy of the Department of Comprehensive Dentistry, University of Texas Health Science Center at San Antonio.)

Box 5-1	**General guidelines for caries management for a high-caries-risk patient, based on PTPM**

Step 1: Plaque control
- Provide prophylactic treatment followed by fluoride application
- See patient regularly to reinforce oral hygiene

Step 2: Treatment of existing caries lesions
- Treat noncavitated lesions as needed
- Restore cavitated lesions and seal surrounding pits and fissures as needed
- Salivary flow measurement to check for dry mouth and treat for xerostomia if needed

Step 3: Protection of surfaces at risk
- Seal all deep pits and fissures
- Apply fluoride varnish to exposed roots

Step 4: Maintenance care for prevention
- Review oral hygiene and dietary habits and advise patient to:
  - Reduce the number of between-meal sweet snacks

❙ Substitute snacks rich in protein
• Advise patient to brush at least twice daily with high fluoride❙delivery toothpaste
• Advise patient to floss once daily
• Provide home treatment and/or other adjunctive therapy:
❙ Use over-the-counter fluoride rinse daily
❙ Chew or suck xylitol-containing gum or candies three times daily
• Recall patient every 3 months to:
❙ Reevaluate current caries risk
❙ Receive fluoride varnish treatment of all teeth
• Obtain bitewing radiographs every 3 to 6 months to check for lesions

Box 5-2	General guidelines for caries management for a moderate-caries-risk patient, based on PTPM

Step 1: Plaque control
• Provide prophylactic treatment followed by fluoride application
• See patient regularly to reinforce oral hygiene

Step 2: Treatment of existing caries lesions
• Treat noncavitated lesions as needed
• Restore cavitated lesions and seal surrounding pits and fissures as needed

Step 3: Protection of surfaces at risk
• Seal all deep pits and fissures
• Apply fluoride varnish to exposed roots

Step 4: Maintenance care for prevention
• Review oral hygiene and dietary habits and advise patient to:
❙ Reduce the number of between-meal sweet snacks
❙ Substitute snacks rich in protein
• Advise patient to brush twice daily with over-thecounter fluoride toothpaste
• Advise patient to floss once daily
• Provide home treatment and/or other adjunctive therapy:
❙ Use over-the-counter fluoride rinse daily
❙ Chew or suck xylitol-containing gum or candies three times daily
• Recall patient every 3 to 6 months to:
❙ Reevaluate current caries risk
❙ Receive fluoride varnish treatment of all teeth
• Obtain bitewing radiographs every 12 to 18 months to check for cavities

Box 5-3	General guidelines for caries management for a low-caries-risk patient, based on PTPM

Step 1: Plaque control

• Continue to reinforce oral hygiene

Step 2: Maintenance care for prevention
• Review oral hygiene habits
• Advise patient to brush twice daily with over-the-counter fluoride toothpaste
• Recall patient every 12 to 24 months to reevaluate current caries risk
• Obtain bitewing radiographs every 24 months to check for caries

Preventive and Noninvasive Treatment Options

The dentist is now in a position to develop the preventive treatment plan based on the identified risk factors associated with the individual patient. However, it may be pertinent to first discuss the concept of "caries balance"[99] (see Fig 5-23) as the ultimate objective of the practitioner, which should be to reduce the risk factors and increase the protective factors that relate to the individual patient. Our current knowledge of the caries process can be used to summarize that caries is:

- A process involving commensal bacteria in plaque
- Dependent on dietary fermentable carbohydrates
- Driven by frequency of eating
- Modified by the inherent nature of the tooth
- Modified by fluoride
- Modified by saliva
- Reversible in its early stages
- Controllable in terms of lesion formation and progression

Thus, it is clinically relevant to consider caries and its progression or regression as an ongoing and often changing balance between pathologic (risk) factors and protective factors[99] (see Fig 5-23). If the pathologic factors outweigh the protective factors, then caries lesions develop and progress. In the reverse situation, caries lesion development is inhibited, and an early lesion is arrested or reversed.

It is logical, from the above summary, to state that the development of caries lesions can be prevented and, if caries lesions form, their progression can be controlled. This can be achieved by changing the local biochemistry of the oral environment to one that reduces the proliferation of cariogenic bacteria, neutralizes plaque acid, and provides the required minerals for lesion remineralization through

alteration in oral hygiene and dietary behavior of the patient, fluoride application, and saliva stimulation.

The following approach may be applied in relation to the risk factors associated with the individual patient to develop a personalized preventive treatment plan for the patient:

- Caries-inactive/caries-controlled patients. These patients present with no new lesions and need only maintenance care, discussed in the following sections, to continue good oral hygiene with a fluoride-containing toothpaste.
- Caries-active patients with relevant risk factors that can potentially be changed. These patients present with lesions. Mechanical plaque control should be improved and consideration given to supplementing fluoride toothpaste with mouthrinses and/or chairside fluoride applications. Where diet investigation shows multiple sugar attacks, advice should be given as to how this might be improved.
- Caries-active patients with some risk factors that cannot be changed. These cases are difficult because the progress that can be made will depend very much on individual circumstances. For instance, if a disability is preventing adequate plaque control, is it possible for a caregiver to help? Social deprivation and level of education may mean that the preventive treatment required is not considered of sufficient importance by the patient, and it may, indeed, be unaffordable. In this case, motivational interviewing may help to increase inherent motivation. A dry mouth is often a permanent condition, and preventive treatments, plaque control including professional tooth cleaning, use of fluoride, dietary modification, and stimulation of salivary flow all may play a role. Saliva substitutes that contain fluoride and other inorganic ions will protect against caries, in addition to providing comfort and relief for dry mouth conditions, providing a protective coating for teeth and oral tissues, and helping to control bad breath.
- Caries-active patients with some risk factors that have not been identified. This is the most frustrating group for the dentist. Something has been missed, so the detective work must continue, and the case must be managed as with the preceding group.

In caries management, choosing the right products can help protect patients. The following products and strategies can be selected in relation to the risk factors associated with the individual patient.

Self-applied preventive measures

Fluoride products

Fluoride dentifrices for plaque control. Regular dentifrices. The most plausible explanation for the decline of caries prevalence is the steady improvement in oral hygiene, which results in regular disturbance and partial removal of dental plaque, combined with a regular daily administration of fluoride, provided via toothpastes.[22,102] A recently published analysis of data derived from a sample of 1,450 preschool children studied in the British National Diet and Nutrition Survey confirmed the significant caries-inhibiting effect of toothbrushing with a fluoride dentifrice twice daily. On a population basis, sugar-containing foods and drinks were not associated with caries experience, unless children brushed once a day or less.[103] Thus, the cornerstone of any preventive strategy for the management of caries is oral self care: twice daily careful cleaning of the teeth with a toothbrush and an effective fluoride toothpaste. Additionally, dental floss should be used, but patients should be instructed carefully, and the frequency of flossing should be recommended individually in relation to oral health needs and the characteristics of the patient: dental floss could be indicated for individuals with closed interdental spaces and interproximal brushes for periodontal patients or in those with open embrasures.[104] Flossing once daily may be recommended for caries prevention, but twice daily flossing may be required if gingivitis needs to be prevented as well.[105] In areas where the water is fluoridated to optimal levels, twice daily careful cleaning of the teeth with fluoride toothpaste is a safe and effective preventive treatment and caries management strategy.[106–108]

Dentifrices with antimicrobials. Some elements combined with fluoride in dentifrices can provide additional benefit and may be considered while advising a patient on the choice of toothpaste. The antimicrobial compound amine (Am) and stannous ions (Sn) have been combined with fluoride (F). Amine has intrinsic antiglycolytic properties, and therefore the antimicrobial effect of amine fluoride (AmF) is stronger as compared with sodium fluoride (NaF).[22,109–111] The stannous ion itself does not affect mineral dissolution, but studies have shown the antimicrobial activity of the SnF_2 complex in vivo.[112,113] Its effect on the inhibition of plaque formation is stronger as compared with other fluoride combinations. AmF/SnF_2 can be effective in low concentrations by slowing down bacterial growth through the inhibition of bacterial metabolism (eg, acid production) Triclosan in

toothpaste has long been demonstrated to provide antibacterial effects by disrupting bacterial cell walls.[114] These products may be more suitable for those patients at high caries risk because of their poor oral hygiene attitude.

High–fluoride content dentifrices. Prescription-strength topical fluorides are often recommended to patients at high caries risk after assessing fluoride exposure of the patient. Lower-level fluoride (eg, in the United States, 1,100 ppm fluoride) dentifrices generally lead to a caries reduction of around 25%88,115; this is insufficient protection for high-risk patients, and therefore alternative therapies are usually needed for these patients. A patient's daily use of a conventional 1,100-ppm fluoride-containing dentifrice may be substituted with a high-fluoride dentifrice (eg, 5,000 ppm). Clinical studies have shown that as much as a 75% reduction in caries incidence may be achieved through use of 5,000-ppm fluoride topical treatments.[115]

Options when plaque control is insufficient. If caries or caries lesion progression is not prevented, the reasons should be carefully examined. A first step in this procedure is to carefully assess the quality of the oral hygiene (Fig 5-27). If hygiene is adequate, it is appropriate to evaluate whether additional risk factors, such as multiple intake episodes of sugar-containing foods and drinks, are present. If so, these risks must be reduced as much as possible. In the meantime, the patient can be helped with professionally applied preventive measures, such as topical applications of concentrated fluoride solutions, gels, or varnishes[100,116] (Tables 5-11 and 5-12). The benefits of topical fluorides are firmly established based on a sizeable body of evidence from randomized controlled trials. The size of the reductions in caries increment in both the permanent and the primary dentitions emphasizes the importance of including topical fluoride delivered through toothpastes, rinses, gels, or varnishes in any caries prevention program.[117,118] Salivary flow can be stimulated by daily use of sugar-free chewing gum. When no additional risks can be identified, the fluoride supply must be intensified, perhaps by adding a third daily fluoride application in the form of additional brushing, a mouthwash, or a tablet (see Tables 5-11 and 5-12). Although fluoride toothpastes in comparison with mouthrinses or gels appear to have a similar degree of effectiveness for the prevention of dental caries in children,[119] topical fluorides (mouthrinses, gels, or varnishes) used in addition to fluoride toothpaste achieve a modest reduction in caries compared with toothpaste used alone.[120] If the daily oral hygiene procedures are inadequate, an attempt should be made to determine whether the problem is due to an inability to use a toothbrush or whether the patient simply is

noncompliant. The dentist or hygienist should apply disclosing solution to the teeth and watch as the patient demonstrates the oral hygiene procedures. If he or she is not able to remove the plaque, the patient should be taught alternative methods. Sometimes a patient can be helped by professionally applied preventive measures. If a patient is able to remove plaque but is not motivated to do so, the dentist/hygienist must try to determine the reasons. The patient may not be convinced of the necessity for thorough plaque removal. This puts the dentist/ hygienist in the realm of behavior modification, a subject not within the scope of this chapter but well discussed by Freeman and Ismail.[121]

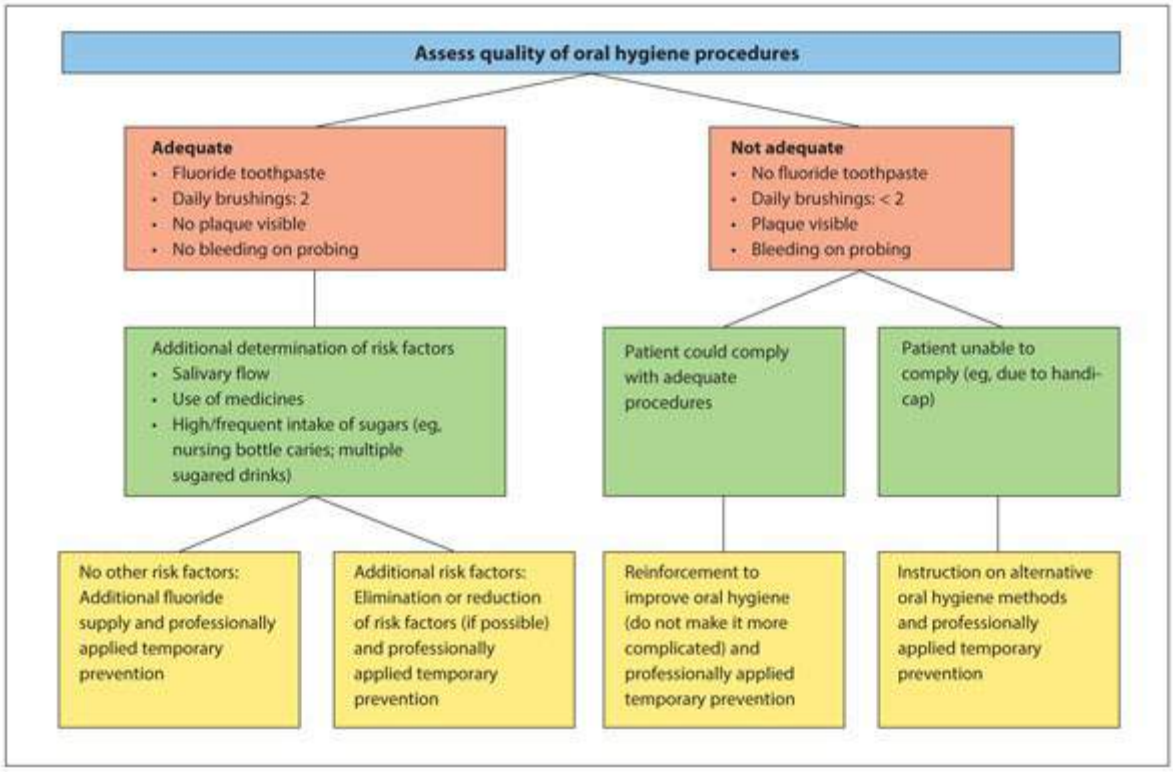


Fig 5-27 Assessment of oral hygiene procedures when caries or caries progression is not prevented.

Table 5-11	**Guidelines for basic preventive advice and the use of recommended preventive products***	
Basic preventive steps for moderate/high-risk patients		**Special needs**
Patient motivation	Emphasis on behavioral change	R, H
Diet counseling	Reduction of fermentable carbohydrate intake and frequency	R, H
	Reduction of soft drink consumption and frequency	

Toothbrushing	Twice daily with fluoride toothpaste (preferably three times a day)	R, H
Flossing	Daily, few times a week	R, H
Sugar-free gum	Two pieces for ≥ 5 minutes three times a day (after each meal preferred)	R, H
Sealants	All at-risk surfaces (sound or noncavitated)	H
Adjunct topical therapies for moderate/high-risk patients		**Special needs**
Home fluoride options:		
Prescription fluoride (5,000 ppm) toothpaste	• Brush at least twice a day (preferably three times a day), including immediately before going to bed • Follow motto "spit, don't rinse" • Apply via tray daily for patients with radiation-induced hyposalivation	R, H
Fluoride rinses	Twice daily/daily/weekly depending on need and produ	R, H
In-office fluoride options:		
Fluoride gel/foams	• 1.23% APF or neutral 2% NaF • 4 minutes, two to four times per year • For root caries, four times over 2 to 4 weeks	R, H
Fluoride varnishes	• Apply to lesions and other surfaces at risk two to four times per year depending on risk	R, H
Potentially helpful adjuncts to home fluoride:		**Special needs**
Xylitol chewing gum	• Xylitol must be listed as first ingredient • Two pieces for ≥ 5 minutes, three times per day (after each meal preferred) for 2 weeks	H
Calcium-based products	Includes paste, toothpaste, mints with/without fluoride; regimen depends on product	H

*Modified from Peters[116] with permission. H–hyposalivation; R–root caries.

Table 5-12	ADA evidence-based recommendations for professionally applied topical fluoride[100]		
Risk group	**< 6 years**	**6–18 years**	**> 18 years**
Low	May not receive additional benefit from topical fluoride	May not receive additional benefit from topical fluoride	May not receive additional benefit from topical fluoride
Medium	Varnish every 6 months	Varnish or fluoride gel application every 6 months	Varnish or fluoride gel application every 6 months
High	Varnish every 3 or 6 months	Varnish every 3 or 6 months	Varnish every 3 or 6 months
		or	or
		Fluoride gel application every 3 or 6 mon	Fluoride gel application every 3 or 6 months

Fluoride gels, mouthrinses, and supplements. The benefits of topical fluorides have been firmly established on a sizeable body of evidence from randomized controlled trials.[122] There is clear evidence of the caries-inhibiting effect of fluoride gel, with the best estimate of the magnitude of this effect, based on 14 placebo-controlled trials, being a 21% reduction in DMFS.[123,124] The most commonly available at-home-use fluoride gels for daily application come in prescription-strength 1.1% NaF. These are recommended for high-risk adults and children 6 years and older and are to be applied once daily over the tooth surfaces using a finger, toothbrush, or individually fitted trays, preferably immediately before bed. The gel should be left in contact with the teeth surfaces for 5 minutes, after which the individual should spit and not rinse. There is little information concerning effects on the primary dentition and on adverse effects, so the prescriber should take responsibility for possible side effects and misuse. Clinical trials have shown that, depending on the strengths and rinsing frequencies, supervised regular use of a fluoride mouthrinse is associated with a clear reduction in caries occurrence in children.[125] Over-the-counter (OTC) mouthrinses are available as 0.05% or 0.02% NaF (in the United States) for daily use. Prescription-strength 0.2% NaF rinses may

be used daily or weekly depending on the caries risk status of the patients. It is recommended that 10 mL of the fluoridated mouthrinse be swished around in the mouth for 1 minute. Clinical trials of both the daily and weekly regimen show an average caries reduction of 30%.[126]

Mouthrinses have also been formulated with acidulated phosphate fluoride (APF), stannous fluoride, ammonium fluoride, and amine fluoride. Acidulation of the solutions may increase the fluoride uptake into the tooth mineral.[22,127,128] Application of an acidulated product causes mild etching of the enamel, which creates micropores for increased fluoride diffusion and increases the surface-reactive area for fluoride. The result is increased uptake and accumulation of fluoride within a caries lesion or healthy tissue. Etching may also cause the transformation of apatite to brushite as an intermediate product. The brushite in turn is readily recrystallized into fluoroapatite when exposed to low concentrations of fluoride (< 1,000 ppm), especially in the presence of a low F/Ca ratio. In the presence of higher fluoride concentrations, however, brushite reacts instead to form calcium fluoride (CaF_2) globules, which are covered by salivary components preventing immediate dissolution. Under acidic challenge, the solubility of the CaF_2 globules increases to release calcium and fluoride in the environment. Therefore, the low pH of APF enhances more rapid fluoride uptake by enamel, while the presence of the orthophosphate prevents enamel dissolution by saturating the immediate environment of the tooth surface with phosphate ions (common ion effect).[22,127,128]

Caution: Although some of these mouthrinses tend to recharge glass-ionomer restorations, the pH of the topical fluoride used to recharge glass-ionomer restorations is important. Acidic topical fluoride solutions, such as APF solutions and other acidified fluoride preparations, degrade glass-ionomer materials, porcelain crowns, and veneers and should be avoided in these patients.[129–131] Resin-modified glass ionomers are more resistant to surface degradation than conventional glass ionomers but still degrade when exposed to acids and orange juice.[123] Resin composites are also degraded by frequent applications of acidic fluoride solutions, producing filler dislodgment and destruction of the filler-resin matrix interface.[132]

Nonfluoride products

Xylitol products. The noncariogenicity of xylitol, demonstrated in clinical studies through its use in chewing gum,[133,134] oral syrup,[135] and gummy bears,[136] is based on the inability of the cariogenic bacteria to metabolize this sugar and produce acid. Xylitol in gums and candies has been shown to have a caries-preventive effect,

which is probably based on stimulation of salivary flow, although an antimicrobial effect cannot be excluded. Xylitol chewing gums can be recommended for saliva stimulation, especially in moderate- and high-caries-risk individuals, and most essentially for patients with hyposali-vation. Two pieces of gum should be chewed for at least 5 minutes three times daily, preferably after each meal. The minimum effective dose for xylitol has not been established, but studies employing daily doses of between 0.9 and 7.0 g have been shown to be effective.[137] Although no minimum frequency of exposure has been established, there are some indications that there is an exposure frequency effect and that at least three exposures daily should be achieved.[137]

Calcium-based dentifrices and mouthrinses. Calcium-based agents, such as nanohydroxyapatite (nHAP), tricalcium phosphate (TCP), calcium sodium phosphosilicate (NovaMin, Nova-Min Technologies), and amorphous calcium phosphate (ACP), have been incorporated into toothpastes to prevent the development of caries and/or aid remineralization/arrest of caries lesions. nHAP-based toothpaste has been commercially available in Japan since the 1980s and was approved as an anticaries agent in 1993 in Japan, following an anticaries field trial in Japanese schoolchildren.[138] A combined fluoride plus functionalized β-TCP is currently being added into single-phase aqueous or nonaqueous topical fluoride formulations (eg, dentifrices, rinses, varnishes, etc).[139,140] In these formulations, functionalization of TCP with specific organic molecules (eg, fumaric acid or sodium lauryl sulfate) provides a barrier that prevents unwanted TCP-fluoride interactions and also provides targeted delivery of TCP when applied to the teeth. Placebo-controlled clinical studies have demonstrated products with functionalized TCP to improve remineralization of white spot lesions.[141,142]

The potential of NovaMin-based toothpaste to prevent demineralization and aid in remineralization of enamel has been demonstrated. When introduced into the oral environment, the material releases sodium, calcium, and phosphate that then can interact with the oral fluids, resulting in the formation of a crystalline hydroxycarbonate apatite layer.[143]

ACP technology is commercially available as toothpaste based on unstabilized ACP, in which a calcium salt and a phosphate salt are delivered separately intraorally via a dual-chamber device or delivered in a product with a low water activity.[144,145] As the salts mix with saliva, they dissolve, releasing calcium and phosphate ions. The mixing of calcium ions with phosphate ions results in the immediate precipitation of ACP or, in the presence of fluoride ions, amorphous

calcium fluoride phosphate (ACFP). In the intraoral environment, these phases (ACP and ACFP) are potentially very unstable and may rapidly transform into more thermodynamically stable, crystalline phases (eg, hydroxyapatite and fluorhydroxyapatite).[144] Clinical studies have demonstrated the ability of ACP-based dentifrices to lower the incidence of root caries[146] and ACP-based mouthrinses to promote remineralization of enamel subsurface lesions.[147] However, there is concern about promotion of dental calculus formation with long-term use.

Dietary management for caries prevention

Considerable evidence exists that between-meal snacks favor development of dental caries.[148] A frequent and prolonged eating pattern, such as snacking with sugary foods between meals, increases the caries risk of an individual[149] because in a frequent-eating condition the pH is seldom allowed to come to neutrality. Consequently, demineralization will outweigh remineralization, and caries will develop and progress. However, it is often not possible to gain the compliance of people when asking them to abstain from snacking. Alternatively, patients can be advised to snack smartly by choosing nonsugary and/ or low-fat snack foods, such as raw vegetables, fresh fruits, or whole-grain crackers or bread with margarine or peanut butter, low-fat (or filled milk) cheese, lean meats, or skim milk. The nature of the foods as well as food combinations influence their effect on the caries process. Thus, the following basic dietary principles may help to reduce the risk for dental caries:

- Eat a diet that is low in sucrose and retentive fermentable carbohydrates.
- Reduce the frequency of eating or drinking fermentable carbohydrates.
- Combine cooked and processed foods with nonacidogenic foods.
- Do not eat cariogenic snacks.
- Include foods of firm or hard texture.
- Choose fats in diet wisely to reduce risk of chronic disease while still benefiting from fat coating on tooth surfaces, which reduces the adherence of plaque.
- Chew sugarless gum (preferably with xylitol) for 15 to 20 minutes after eating to increase salivary benefits.
- Combine and sequence foods to encourage chewing and saliva production.
- Only eat sweets with noncariogenic sweeteners.

Professionally applied preventive measures

Varnishes, gels, and foams

Professionally applied topical fluorides in gels or varnishes have been shown to be effective in many clinical trials.[119,120,122,150] The benefits of topical fluorides are firmly established based on a sizeable body of evidence from randomized controlled trials. The size of the reductions in caries occurrence in both the permanent and the primary dentitions emphasizes the importance of including topical fluoride delivered through toothpastes, rinses, gels, or varnishes in any caries preventive program.[117] A substantial caries-inhibiting effect of fluoride varnish in both the permanent and the primary dentitions based largely on trials has been reported.[151–153] Fluoride varnish treatment twice a year at 6-month intervals prevented proximal caries in adolescents as much as 69% in high-, 66% in medium-, and 20% in low-risk individuals.[152] Table 5-12 summarizes the American Dental Association (ADA) Council on Scientific Affairs clinical recommendations for professionally applied topical fluoride.[100] Fluoride varnishes contain a high level of fluoride (5% NaF, 22,600 ppm F), and its application on tooth surfaces results in formation of a deposit of calcium fluoride globules that can act as a reservoir for the slow release of fluoride and calcium over time under acid challenge.

Several excellent studies confirm that fluoride varnish application is effective in reversing and arresting active enamel lesions and therefore reduces the need for restorative intervention.[154–160] Peyron et al[161] concluded that fluoride varnish application has a definite cariostatic effect on proximal caries. Following isolation of each quadrant of the dentition with cotton rolls, fluoride varnish is applied to the lesions and other surfaces at risk about two to four times per year, depending on the caries risk of the individual patient. The varnish can be dragged onto the proximal surfaces with dental floss using forward and backward strokes. Fluoride gels and foams, such as 1.23% acidulated phosphate fluoride or neutral 2% NaF, are applied with custom-made trays for 4 minutes, two to four times per year or, for root caries, four times over 2 to 4 weeks.

Pit and fissure sealants

Fissures are more susceptible to caries attack than smooth surfaces because fissure anatomy favors plaque maturation and retention.[162] However, lesions can be prevented, and active white spot lesions can be arrested by brushing alone.[62] Sealing

is a recommended procedure to prevent caries of the occlusal surfaces of permanent molars,[163] but it is only required if occlusal surfaces are not being cleaned. Pit and fissure sealants have been used successfully for many years to prevent caries by preventing the growth of bacteria that promote decay in pits and fissures in teeth. Sealants may be indicated for children and adults who, because of poor oral hygiene, may be at moderate or high risk of developing dental caries, have active white spot lesions (limited to enamel of pits and fissures), or have existing pits and fissures that are anatomically susceptible to caries.

Currently there are two main types of pit and fissure sealant materials available: resin-based sealants/composites and glass-ionomer cements. The effectiveness of resin-based sealants has been demonstrated in many studies.[164,165] Glass-ionomer cements contain fluoride, and it has been suggested that glass-ionomer sealants, through their fluoride release, can prevent the development of caries even after the visible loss of sealant material.[166] Pits and fissures in partially erupted molars where moisture control for placement of a conventional sealant is difficult can be temporarily protected with a recently developed moisture-tolerant material. This is a light-cured glass-ionomer material in varnish form that is recommended for site-specific, protective coating of enamel and dentin tooth surfaces, especially as a protective coating for at-risk surfaces.

Noninvasive Options for Treatment of Existing Lesions

Treatment of noncavitated lesions

Lesion arrest (remineralization therapy)

Active noncavitated caries lesions on occlusal, proximal, and nonproximal coronal smooth surfaces or root surfaces can be arrested. The choice of treatment for active noncavitated lesions on proximal coronal surfaces depends on the caries risk status of the patient as well as the depth of the lesion (see Fig 5-14). Arrest of active caries lesions can be achieved successfully with a combination of improved oral hygiene and application of topical fluoride agents, such as varnishes, gels, or foams. Application of the agent can be repeated every 3 months until caries activity is under control. This regimen should be employed with good oral hygiene and a daily use of high fluoride–containing toothpaste/or mouthrinse.

The average speed of progression of caries lesions on different surfaces has been

determined.[66,167] On proximal and free, smooth areas, caries proceeds slowly. It is thus reasonable to postpone restorative intervention. A procedure whereby the patient is examined regularly ("watchful waiting") creates the opportunity for arrest and remineralization. "Watchful waiting" implies that nothing is done, but in fact it is based on the intellectual decision not to restore because of knowledge of the caries process. The dentist is performing active preventive treatment by helping the patient to improve oral hygiene, by applying fluoride, and by encouraging the patient to modify his or her diet. These measures should always be carried out when active noncavitated lesions exist.

Occlusal sealants

The choice of management of an initial noncavitated caries lesion depends on the caries risk status of the individual patient and the location of the lesion in the dentition. When active fissure caries has been diagnosed or if a high risk has been established, and fissures have susceptible morphologic characteristics, sealants may be indicated (see Fig 5-24). After acid etching (for a lightly filled resin fissure sealant or a flowable resin composite) or conditioning (for a glass-ionomer cement), the sealant material is used to penetrate the fissures, rendering it self-cleansing to prevent plaque accumulation. This is especially important during the period of tooth eruption, although the application of sealants in suspect fissures is also advisable in older patients with high caries risk.[168] There are several advantages of fissure sealants: No irreversible interventions are necessary, active dentin lesions inadvertently covered by the resin do not progress further, and the possible development of new lesions in other sites of the fissure is prevented. Sealants have been used successfully for many years.[169] Concern has been expressed about placement of sealants over undetected dentin caries lesions. However, there is ample evidence that caries lesions do not progress as long as the fissure remains sealed.[170–175] Sealed, radiographically evident caries lesions have been shown in one study not to progress over a 10-year period.[174,175]

Proximal sealants

The concept of arresting caries by applying a resin barrier to the surface, as in occlusal fissure sealing, has also been transferred to proximal surfaces. Active caries lesions on proximal surfaces have been treated by application of either adhesives or fissure sealants after temporary tooth separation.[176–180] Clinical trials

suggest that novel techniques for sealing proximal lesions show promise.[176–180] However, proximal sealing techniques are as complex to apply and time-consuming as proximal restorations.

Lesion infiltration

Lesion infiltration technique is an emerging alternative method of treating active noncavitated lesions extending radiographically into inner enamel or the outer third of dentin that are located on the nonproximal and proximal coronal smooth surfaces.[181] In contrast to sealing of fissure caries, where the protective resin layer is established on the tooth surface, caries infiltration occludes lesion pores with low-viscosity light-curing resins in order to create a diffusion barrier and hence arrest caries progression. Following etching of the lesion with hydrochloric acid and achievement of a dried surface using ethanol and air drying, the low-viscosity infiltrant is applied on the lesion surface in two stages (Fig 5-28). In the first application, the infiltrant is allowed to infiltrate the lesion for 3 minutes and then is light cured. In the second application, the infiltrant is allowed to infiltrate for only 1 minute and then is light cured. The infiltration technique is based on the fact that if the lesion body is sufficiently well infiltrated with resin, lesion progression is significantly reduced in a cariogenic environment; moreover, infiltration might stabilize the fragile lesion structure.[181,182] However, the current infiltrant is radiolucent, making it difficult to determine the depth of infiltration or identify an already infiltrated lesion in a radiograph.

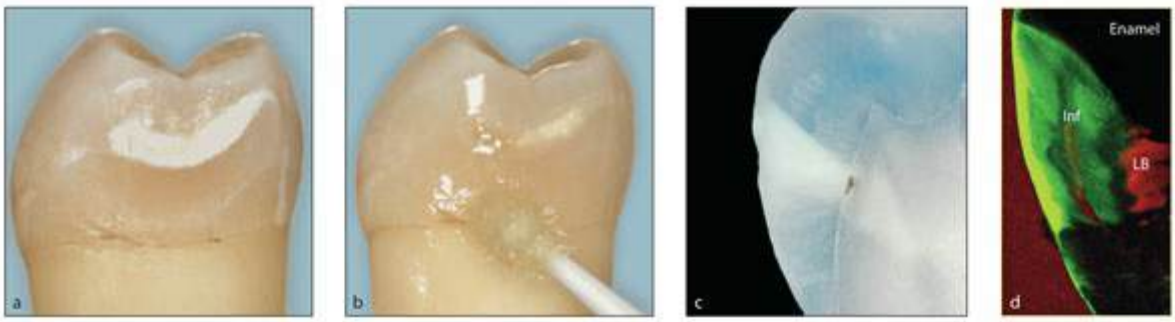


Fig 5-28 Caries infiltration. *(a)* Active noncavitated caries lesions (white spot lesion). *(b)* After application of an infiltrant, the lesion loses its whitish appearance. *(c)* The cross section of a noncavitated proximal caries lesion. *(d)* Confocal microscopic image of an infiltrated lesion. The infiltrant (Inf) is shown in green, whereas the remaining pores within the lesion are shown in red. The infiltrant almost completely penetrates the lesion body (LB) and thereby acts as a diffusion barrier inside the lesion. (Courtesy of Sebastian Paris and Hendrik Meyer-Lückel, Aachen, Germany.)

Treatment of cavitated lesions

Caries lesions may be detected at a relatively late point of development. This may be because of difficulties in detecting the early lesion, or it may be because the patient did not visit the dentist early in the lesion process. Although this may suggest the value of a rigorous, invasive approach, it is preferable to select a treatment option that is as conservative as possible because of the iatrogenic damage that may occur during preparation. Lussi and Gygax[183] showed that during the preparation of a proximal surface, the neighboring surface was damaged 100% of the time, despite very careful operating procedures. Iatrogenic preparation damage represents a dental health problem, because the damage increases the caries progression and the perceived need for restorative therapy of the adjacent teeth.[184]

Although lesions that are cavitated are treated traditionally by preparation and restoration, clinical observations suggest that caries lesions can be arrested at any stage of lesion development (ie, even at the cavitation stage if plaque-free conditions are maintained).[79] Thus, a preventive treatment approach is often successful, especially when the lesion is in a free, smooth surface. When a lesion is present in an occlusal or proximal surface, it will often be difficult to arrest lesion progression because of the difficulty of removing plaque. Before invasive procedures are initiated, noninvasive options must be explored and preventive measures taken.

Several authors have shown that when an occlusal lesion has cavitated, the dentin is always involved in the process.[55,56] Moreover, these lesions, mostly detectable on radiographs, contain many microorganisms and are therefore considered active.[57,58,64,65] Measures directed at a thorough removal of plaque are ineffective on the occlusal surface because the bristles of the toothbrush cannot get into the undermined cavity. Proximal cavitations are also difficult to reach. Dental floss will skim the surface but will not achieve access to the cavitated area. However, where there are cavitated areas on free, smooth surfaces, the situation is different. Those areas are easily reached by the toothbrush but may be difficult to clean due to undermining of the enamel. Removal of the overhanging enamel margins must be considered to aid in keeping the whole area free of plaque. Cavities in these surfaces, cleaned twice daily with a fluoride toothpaste, can be arrested and converted into leathery or hard lesions[79,185] (see Figs 5-20 and 5-21). When the activity of highly infected caries lesions is decreased and finally arrested, the carious layers contain few bacteria that can be cultivated.[186–189] Although occlusal or proximal caries lesions cannot be approached by preventive measures alone, in the primary dentition this method can be successful (Fig 5-29). Therefore,

undermined enamel margins should be eliminated so that when plaque is removed, fluoride can be applied easily to the carious dentin. Under ideal conditions, carious dentitions can be managed so that caries is arrested and demineralization and remineralization are in equilibrium (Fig 5-30; see also Fig 5-29).



Fig 5-29*(a and b)* Arrested caries lesions in the primary dentition.

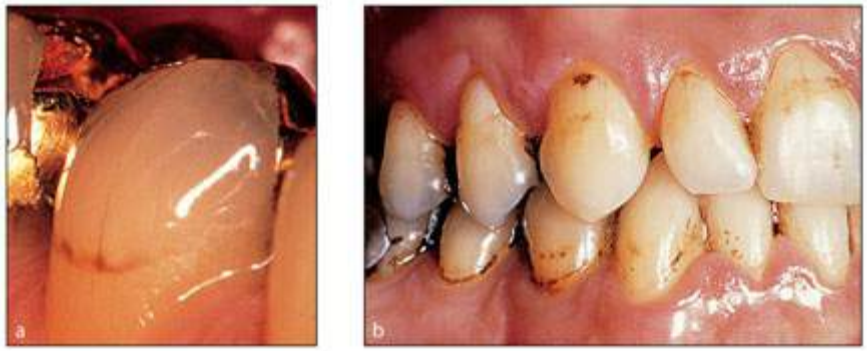


Fig 5-30*(a and b)* Arrested lesions seen as brownish "scar tissue."

Operative and Minimally Invasive Treatment Options

Restoration of noncavitated (initial) lesions

Bitewing radiographs are the most effective method of detecting proximal lesions, but correlation with surface integrity is poor. The algorithm shown in Fig 5-14 demonstrates that caries lesions on proximal surfaces are considered differently when it comes to restorative treatment. In addition to the caries risk status of the patient, the depth of the demineralization plays a role in the choice of treatment. Lesions that extend on the radiograph slightly past the dentinoenamel junction (D1 lesions as classified in Fig 5-14) are selected by some dentists for restorative treatment. Such a decision is based on the fact that radiographs have been shown to considerably underestimate the histologic size of the lesion,[71] and the possible

prediction of the cavitation of proximal lesions is based on the radiographic depth and caries activity of the patient.[72] However, in contemporary populations, these deep proximal lesions are rarely cavitated. They should be treated preventively as described previously and reassessed clinically and radiographically at an appropriate recall interval, depending on the patient's caries risk status. Teeth separation using orthodontic elastic separators, allowing impression/inspection, may serve as an effective diagnostic tool for assessing surface integrity of proximal lesions.[72]

Restoration of cavitated lesions

Cavitated caries lesions are candidates for traditional restorative treatment, such as resin composite, resin-modified glass ionomer, or a sandwich technique combining resin-modified glass ionomer and resin composite or amalgam. It is necessary to have well-defined criteria for the decision to restore a tooth due to caries. The most important reason for placing a restoration is to aid plaque control. Elderton and Mjör[190] formulated the following indications for restorative treatment:

- The cavitated tooth is sensitive to hot, cold, sweetness, etc.
- Occlusal and proximal lesions extend deep into dentin (and cannot be reached by the toothbrush).
- The pulp is endangered.
- Previous attempts to arrest the lesion have failed, and there is evidence that the lesion is progressing (such evidence usually requires an observation period of months or years).
- The patient's ability to provide effective home care is impaired.
- Drifting is likely to occur through loss of proximal contact.
- The tooth has an unesthetic appearance.

Treatment will be directed in such a way that infected dental tissue is removed and the remaining cavity is adapted so that the restorative material can be optimally placed. The particular preparation methods and restorative procedures are discussed in subsequent chapters.

Preventive resin restorations and preventive amalgam restorations

Restoration of the carious surface should be accompanied with sealing of

surrounding pits and fissures. Sealed restorations placed directly over frankly cavitated lesions can arrest lesion progression. Sealing of restorations, therefore, conserves sound tooth structure, protects the margins of restorations, and prevents recurrent caries (secondary caries).[175]

Interim therapeutic restorations

Interim therapeutic restorations (ITRs) are utilized as part of comprehensive caries management when caries control is necessary prior to placement of definitive restorations. ITRs may be employed in patients with multiple caries lesions to temporarily restore the lesions prior to definitive restoration of the teeth.[191] It should be applied to children experiencing early childhood caries,[192] in cases of rampant caries, in very high caries risk cases, to uncooperative patients, to patients with special health care needs,[193] or when traditional cavity preparation and/or placement of traditional dental restorations is not feasible and needs to be postponed.[191–198] It is also used when there is need for stepwise excavation in children or adults with multiple open caries lesions prior to definitive restoration of the teeth.[199] ITRs may be used to reduce the levels of cariogenic bacteria in the oral cavity[200,201] and make the tooth surfaces more cleansable prior to placing definitive restorations.

The ITR procedure involves removal of caries using hand or low-speed rotary instruments, with caution not to expose the pulp, followed by restoration of the tooth with an adhesive restorative material, such as conventional or resin-modified glass-ionomer cement.[202] It is applicable mainly to single-surface or small two-surface restorations, where the greatest success has been achieved.[203] Maximum caries removal from the periphery of the lesion will help to minimize leakage of the restoration, while inadequate cavity preparation with subsequent lack of retention and insufficient bulk can lead to failure.[203] However, ITRs must be accompanied with maintenance care including oral hygiene instruction, topical fluoride application, and interviews to motivate and educate patients and modify their habits and risk factors during the time that they have transitional restorations.

Atraumatic restorative treatment

Atraumatic restorative treatment (ART) is a procedure based on excavation of carious tooth tissues using hand instruments alone. Thus, it eliminates the need for anesthesia and use of expensive equipment and restores the cavity with glass ionomer, an adhesive material that bonds to the tooth structure and releases

fluoride.[196,204] Some suggest that it may aid in remineralization of tooth structure. Removing carious tooth tissue with hand instruments alone and restoring the cavity with an adhesive material will conserve as much tooth structure as possible and prevent further decay. In addition to its use as a restorative material, glass ionomers can be applied in the very early stages of caries development. The glass ionomer adheres to the tooth and halts or slows the progression of lesions, probably because it slowly releases fluoride.[196] However, ART must be accompanied with maintenance care with oral hygiene instruction and topical fluorides to improve the treatment outcome, especially in high-caries-risk populations.

ART is minimally invasive, making it highly acceptable to patients. It is recommended as a definitive restoration for use in children as well as in fearful adults. It also serves as a definitive restorative option for special groups in the community, such as the physically or mentally handicapped, people living in nursing homes, and the home-bound elderly.[198,204]

ART is considered as a preventive and definitive treatment modality for caries in communities with little or no access to traditional dental care. ART also has its place in the industrialized world as a minimal intervention, minimal invasion, and minimal cavity preparation for caries lesions. The World Health Organization actively promotes ART as a viable approach to meet the need for treatment of dental caries. ART is now considered as part of a total package of oral health care that is based on a philosophy of promoting health and preventing disease.[204]

Maintenance Care

A continuing care program is essential in PCCM because it creates the opportunity to engage the patients in the treatment of caries, and a high level of compliance is required for its success. Thus, as part of the initial management of caries in an individual patient, personalized oral health maintenance care should be included in the treatment planning. Maintenance care should be offered to all patients visiting the dental clinic. It is important to realize that the caries risk status of an individual can change over time. Von der Fehr et al[205] and Koch[206] have shown that normal individuals can turn into individuals with high caries risk. Within a short period of time, numerous initial caries lesions can develop when oral hygiene is withdrawn and a sugar-rich diet is offered.[205,206] However, when effective oral hygiene and application of topical fluoride are instituted and a normal, less sugar-rich diet is consumed, reversal of the situation seems possible. Caries risk assessment is

inevitably a "best guess" based on available evidence. Thus, recall and reassessment are important to review the original decision.

Recall and reassessment visits

The establishment of a recall visit should be a component of the maintenance plan to prevent future disease occurrence. A recall system is a means of establishing a continuing care regimen that provides opportunities to reassess and monitor the oral health of patients and to inform future treatment planning.[207] There is some evidence that recall visits, irrespective of the frequency, have a positive impact in terms of the preservation of the dentition.[208] However, the main objectives of the recall system are the following:

- Prevent new disease occurrence.
- Facilitate early detection of any newly developed disease.
- Assess the status of a previously diagnosed and treated disease.
- Enable the clinician or the patient to consider altering the treatment regimen to obtain a more favorable outcome.
- Provide evidence for future clinical governance and quality improvement services.
- Create opportunities to advise, reassess, and reinforce the appropriateness of previous advice.
- Monitor patient compliance with previous advice and treatment.
- Encourage patient behavior that will improve and maintain their oral and general health.
- Create the opportunity, where possible, of using motivational interviewing to increase the inherent motivation of the patient.

The United Kingdom National Institute for Clinical Excellence (NICE) issued a guidance document in 2004 recommending that the individual patient's caries risk status should inform his or her recall interval[209] (Box 5-5).The recommendations cover risk factors such as caries incidence and restorations; periodontal health and tooth loss; patients' well-being, general health and preventive habits; and pain and anxiety. The information collected during the caries risk assessment should be used to determine a personalized variable time interval to assess, reassess, and monitor the oral health and caries status of patients, based on the clinical judgment and

expertise of the dental team and discussions with the patient. The personalized interval should vary from 3 to 24 months, according to risk (see Boxes 5-1 to 5-3). The recommended recall interval should be discussed with the patient to seek his or her agreement to facilitate compliance with the visits. This interval should be reviewed at the next oral health visit for possible alteration, depending on the prevailing caries risk status and patient compliance; hence, it is important to inform patients that their recommended recall interval may vary over time. Furthermore, the patient's ability or desire to visit the dentist at the recommended interval and the financial costs to the patient of having the oral health review and any subsequent treatments should be discussed. The following key components are recommended at the recall visits:

Box 5-4	NICE[209] recommended method for determining a personalized recall interval

Step 1: Consider the patient's age, which sets the range of recall intervals.
- For children and teenagers under 18 years, the range is 3 to 12 months.
- For adults 18 years or older, the range is 3 to 24 months.

Step 2: Consider modifying factors in light of the patient's medical, social, and dental histories and findings of the clinical examination.

Step 3: Integrate all diagnostic and prognostic information, considering advice from other members of the dental team where appropriate. Use clinical judgment to determine the interval for the next oral health review.

Step 4: Discuss the recommended interval with the patient and record the agreed-upon interval or any reason for disagreement.

Step 5: At the next oral health review, consider whether the interval was appropriate. Adjust the interval depending on the patient's ability to maintain oral health between reviews.

- Discussion between the dental team and the patient (and/or his or her parent, guardian, or caregiver) concerning, where appropriate, the effects of oral hygiene, diet, fluoride use, tobacco, and alcohol on the oral health of the individual patient.
- Discussion on the risk factors that may influence the patient's oral health and their implications to future caries occurrence or failure of the treated lesions.
- General preventive advice on frequency of brushing and the use of fluoride

delivery agents; frequency of flossing[116] (see Table 5-11); and dietary advice, such as reduction of the number of between-meal sweet snacks and substitution with snacks rich in protein.

- Provision of the patient with self-applied (at-home use) caries-preventive agents, such as OTC fluoride rinses, and agents for salivary stimulation, such as xylitol-containing gum or candies[116] (see Table 5-11).
- Establishment of an interval for bitewing radiographs to check for lesions, especially proximal lesions. The current recommendation based on the evidence is that the interval should depend on the caries risk status of the individual patient and should be every 3 to 6 months for patients at high caries risk, every 12 to 18 months for moderate caries risk, and every 24 months for low caries risk.
- Recommendation of the next recall visit and the interval of the visit (see Box 5-4).

The advice is tailored to eliminate or at least minimize the risk factors and increase the protective factors (see Fig 5-23).

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